Analysis II

MATH 255 · Scribed Course Notes

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1 Course Overview and Assignment Results

Professor: Jérôme Vétois

Scribe: Yitian Zhang

Email: yit dot zha3 at gmail dot com

Note 1. The abbreviations used in the manuscript have been expanded in this edition.

Note 2. I left out about 1/3 of the proofs.

Note 3. A good amount of interesting results are given out as assignment questions, which I will list them below.

Assignment 1, Question 2. Parallelogram Law + A norm is induced by an inner product

$$\Longleftrightarrow \text{Parallelogram Law holds.}$$

Question 4. Given (X,d) and $Y\subseteq X$. $A\subseteq Y$ is open (closed) in $(Y,d)\Longleftrightarrow \exists\,\widetilde{A}$ open (closed) in (X,d) such that

$$A=\widetilde{A}\cap Y.$$

Question 5. Strong equivalence of metrics, induced topologies, the equivalence and convergence of p-norms on $\mathbb{R}^d$.

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Assignment 2, Question 2.

$$\bigcup_{i\in I}\overline{A_i}\subseteq\overline{\bigcup_{i\in I}A_i},\qquad \overline{\bigcap_{i\in I}A_i}\subseteq\bigcap_{i\in I}\overline{A_i},\qquad \left(\bigcap_{i\in I}A_i\right)^\circ\subseteq\bigcap_{i\in I}A_i^\circ,$$
$$\bigcup_{i\in I}A_i^\circ\subseteq\left(\bigcup_{i\in I}A_i\right)^\circ;\qquad \overline{\bigcup_{i=1}^n A_i}=\bigcup_{i=1}^n\overline{A_i},\qquad \left(\bigcap_{i=1}^n A_i\right)^\circ=\bigcap_{i=1}^n A_i^\circ.$$

Question 3. If $A\subseteq(X,d)$ is closed, then

$$(\partial A)^\circ=\phi,\qquad \partial(\partial A)=\partial A.$$

Question 4.

$$\{f\in C([a,b])\mid f(a)=f(b)\}\quad\text{is closed in }(C([a,b]),\|\cdot\|_\infty),$$
$$\{f\in C([a,b])\mid f>0\}\quad\text{is open in }(C([a,b]),\|\cdot\|_\infty),$$
$$\mathcal{D}^1([a,b])=\{\text{differentiable functions}\}\quad\text{is neither closed nor open in }(C([a,b]),\|\cdot\|_\infty).$$

Question 5. If (X,d) is separable (that is, $\exists$ countable A such that $\overline{A}=X$ (A is dense in (X,d))), then (Y,d) is separable, $Y\subseteq X$.

Assignment 3, Question 2.

$$\mathcal{B}(I):=\{\text{Bounded functions}\},\qquad \|f\|_\infty=\sup_{x\in I}|f(x)|,\quad\forall f\in\mathcal{B}(I).$$

$I \subseteq \mathbb{R}$ an interval. $(\mathcal{B}(I), \|\cdot\|_\infty)$ is Banach.

Question 3. If A_k is a decreasing sequence ($A_{k+1} \subseteq A_k$) of nonempty compact sets, then

$$\bigcap_{k=1}^{\infty} A_k \neq \phi \qquad (\text{in } (X,d)).$$

If A_k is a decreasing sequence of nonempty closed sets in a complete metric space (X,d) such that

$$\operatorname{diam} A_k := \sup_{(x,y)\in A_k} d(x,y) \xrightarrow{k\to\infty} 0,$$

then $\exists x_0 \in X$ such that

$$\bigcap_{k=1}^{\infty} A_k = \{x_0\}.$$

Question 4. Every compact metric space is separable.

Question 5. $(\mathbb{R}^2, \|\cdot\|_2)$, $(\mathbb{R}, |\cdot|)$, $f : \mathbb{R} \to \mathbb{R}$,

$$A_f := \{(x,y) \in \mathbb{R}^2 \mid f(x) \leq y\}, \qquad B_f := \{(x,y) \in \mathbb{R}^2 \mid f(x) \geq y\}.$$

Then A_f is closed in $(\mathbb{R}^2, \|\cdot\|_2)$ if and only if f is lower semicontinuous, and B_f is closed if and only if f is upper semicontinuous. In particular, both sets are closed when f is continuous.

Assignment 4, Question 1. A linear homomorphism between $(X, \|\cdot\|_X)$, $(Y, \|\cdot\|_Y)$ is continuous $\iff$ it is bounded, that is, $\exists C > 0$ such that

$$\|f(x)\|_Y \leq C\|x\|_X, \qquad \forall x \in X$$

$\iff$ its operator norm

$$\|f\|_{X,Y} := \sup_{\|x\|_X=1} \|f(x)\|_Y < \infty.$$

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Consequently, f is continuous $\iff$ f is Lipschitz.

Question 2. Every linear homomorphism with a finite-dimensional domain is continuous (Lipschitz).

All norms are equivalent on a finite-dimensional normed vector space.

Question 3. Convergence of Neumann Series in $(M_{2\times 2}(\mathbb{R}), \|\cdot\|_{2,2})$, where

$$\|A\|_{2,2} = \sup_{\|x\|_2=1} \|Ax\|_2 : \qquad \|A\|_{2,2} < 1 \implies \sum_{k=0}^{\infty} A^k \text{ converges and equals } (\mathrm{Id} - A)^{-1}.$$

Questions 4 and 5. Nontrivial examples of uniform continuity and the Banach fixed point theorem.

Question 6. If (X,d) is compact, $f : X \to X$ such that

$$\forall x, y \in X, x \neq y \implies d(f(x), f(y)) < d(x,y),$$

then f admits a unique fixed point in X. (Surprisingly, this does not use the Banach fixed point theorem!)

Assignment 5, Question 1. Absolutely Convergence Criterion for Banachness.

Question 2. Cauchy Condensation Test: If a_k is a positive decreasing sequence, then

$$\sum_{k=1}^{\infty} a_k \text{ converges} \quad \iff \quad \sum_{k=1}^{\infty} 2^k a_{2^k} \text{ converges.}$$

Corollary.

$$\sum_{k=2}^{\infty} k^{-\alpha} (\ln k)^{-\beta} \text{ converges} \quad \iff \quad \alpha > 1, \text{ or } (\alpha = 1, \beta > 1).$$

Question 3. Bertrand's convergence test: For a_k strictly positive,

$$\text{if } \limsup_{k\to\infty} \left(k \ln k \left(1 - \frac{1}{k} - \frac{a_{k+1}}{a_k} \right) \right) < 1, \text{ then } \sum_{k=1}^{\infty} a_k \text{ diverges.}$$

$$\text{if } \liminf_{k\to\infty} \left(k \ln k \left(1 - \frac{1}{k} - \frac{a_{k+1}}{a_k} \right) \right) > 1, \text{ then } \sum_{k=1}^{\infty} a_k \text{ converges.}$$

Question 4. Hölder's inequality in ℓ^p spaces.

Question 5. Abel's theorem: If $\sum_{k=0}^{\infty} a_k$ converges in $(\mathbb{R}, |\cdot|)$, then

$$\lim_{x\to 1^-} \left(\sum_{k=0}^{\infty} a_k x^k \right) = \sum_{k=0}^{\infty} a_k.$$

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Assignment 6, Question 1. Let $a, b, s \in \mathbb{R}$, $a < s < b$, $\alpha : [a, b] \to \mathbb{R}$ increasing, $f : [a, b] \to \mathbb{R}$ bounded on $[a, b]$. Then

$$f \in \mathcal{R}([a, b], \alpha) \quad \iff \quad f \in \mathcal{R}([a, s], \alpha) \cap \mathcal{R}([s, b], \alpha).$$

If this is the case, then

$$\int_a^b f \, d\alpha = \int_a^s f \, d\alpha + \int_s^b f \, d\alpha.$$

Question 2. Let $a, b, s, \alpha_a, \alpha_b, \alpha_s \in \mathbb{R}$, $a < s < b$, $\alpha_a \le \alpha_s \le \alpha_b$, $f : [a, b] \to \mathbb{R}$ bounded,

$$\alpha : [a, b] \to \mathbb{R} \quad \text{with} \quad \alpha(x) = \begin{cases} \alpha_a, & \text{if } x < s, \\ \alpha_s, & \text{if } x = s, \\ \alpha_b, & \text{if } x > s. \end{cases}$$

Then

$$f \in \mathcal{R}([a, b], \alpha) \quad \iff \quad f \text{ is } \begin{cases} \text{continuous from the right at } s, & \text{if } \alpha_s = \alpha_a, \\ \text{continuous at } s, & \text{if } \alpha_a < \alpha_s < \alpha_b, \\ \text{continuous from the left at } s, & \text{if } \alpha_s = \alpha_b. \end{cases}$$

If such, then

$$\int_a^b f \, d\alpha = f(s)(\alpha_b - \alpha_a).$$

Question 3. Let $a, b \in \mathbb{R}$, $a < b$, $\alpha : [a, b] \to \mathbb{R}$ increasing on $[a, b]$, and $f : [a, b] \to \mathbb{R}$ bounded. Then if α is differentiable with $\alpha' \in \mathcal{R}([a, b])$, then

1. For $M = \sup\{|f(x)| \mid a \le x \le b\}$, any partition P of $[a,b]$,
$$|U(P,f,\alpha) - U(P,f\alpha')| \le M(U(P,\alpha') - L(P,\alpha')),$$
$$|L(P,f,\alpha) - L(P,f\alpha')| \le M(U(P,\alpha') - L(P,\alpha')).$$
2. By 1, $f \in \mathcal{R}([a,b],\alpha) \iff f\alpha' \in \mathcal{R}([a,b])$.

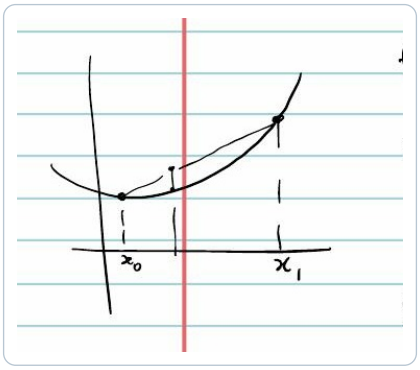
If this is the case, then
$$\int_a^b f\,d\alpha = \int_a^b f\alpha'\,dx.$$

Question 5. Hölder's inequality for Riemann–Stieltjes integral. Let $p \in [1,\infty]$, $a,b \in \mathbb{R}$ such that $a < b$, $\alpha : [a,b] \to \mathbb{R}$ increasing on $[a,b]$. Then
$$\|fg\|_{1,\alpha} \le \|f\|_{p,\alpha}\,\|g\|_{p',\alpha}, \qquad \forall f,g \in \mathcal{R}([a,b],\alpha),$$
with p' being the Hölder conjugate of p:
$$\frac{1}{p} + \frac{1}{p'} = 1, \quad \text{and}$$

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$$\|f\|_{p,\alpha} = \begin{cases} \left(\int_a^b |f|^p\,d\alpha\right)^{1/p}, & \text{if } 1 \le p < \infty, \\ \sup\limits_{x\in[a,b]} |f(x)|, & \text{if } p = \infty. \end{cases}$$

Question 6. Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ twice differentiable on I, $f''(x) \ge 0$, $\forall x \in I$, then f is convex on I, that is, $\forall t \in [0,1]$, $x_0, x_1 \in I$,
$$f((1-t)x_0 + tx_1) \le (1-t)f(x_0) + tf(x_1).$$



Proof via Taylor's theorem with Lagrange's form of remainder.

Question 7. An example using Taylor's theorem with integral form of remainder.

Question 8. An example using the integral comparison test.

Question 9. An example of improper Riemann–Stieltjes integral computation.

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2 Metric, Normed, and Inner Product Spaces

2.1 Metric Spaces

Definition

For a set X, a function

$$d : X \times X \longrightarrow \mathbb{R}, \qquad (x, y) \longmapsto d(x, y),$$

satisfying

1. positive definiteness:

$$\forall x, y \in X, \quad d(x, y) \geq 0, \quad \text{and} \quad d(x, y) = 0 \iff x = y;$$

2. Symmetry:

$$d(x, y) = d(y, x), \qquad \forall x, y \in X;$$

3. $\triangle$-inequality:

$$d(x, z) \leq d(x, y) + d(y, z), \qquad \forall x, y, z \in X,$$

is called a distance function / metric on X. □

Definition

A set equipped with a metric is called a metric space. □

Example

For every $n \in \mathbb{N}$, $p \in [1, \infty]$, define

$$d_p : \mathbb{R}^n \times \mathbb{R}^n \longrightarrow [0, \infty),$$

$$(x, y) \longmapsto \begin{cases} \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p}, & 1 \leq p < \infty, \\ \max\limits_{1 \leq i \leq n} |x_i - y_i|, & p = \infty. \end{cases}$$

Note that this is the ℓ^p norm, also denoted by $\|x - y\|_p$, or simply $\|x - y\|$ if $p = 2$, which is the Euclidean norm / metric. □

Note. We will prove that d_p is a metric on $\mathbb{R}^n$.

Remark. d_p fails to be a norm if $p \in (0, 1)$:

$$d_p((1, 0, \ldots, 0), (0, 1, 0, \ldots, 0)) = (1^p + 1^p)^{1/p} = 2^{1/p} > 2,$$

$$d_p((1, 0, \ldots, 0), (0, \ldots, 0)) + d_p((0, 1, 0, \ldots, 0), (0, \ldots, 0)) = 1 + 1 = 2.$$

The triangle inequality fails.

Example

$$d : \mathbb{C} \times \mathbb{C} \longrightarrow [0, \infty), \qquad (z_1, z_2) \longmapsto \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} = |z_1 - z_2|,$$

where $z_1 = (x_1, y_1)$, $z_2 = (x_2, y_2)$. □

Example

$$d : X \times X \longrightarrow \mathbb{R}$$

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$$(x, y) \longmapsto \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Discrete metric on X. □

2.2 Normed Vector Spaces

Definition

Given a vector space X over $\mathbb{R}$ or $\mathbb{C}$, a function $\|\cdot\| : X \to \mathbb{R}$ satisfying

1. $\forall x \in X$, $\|x\| \geq 0$, and $\|x\| = 0 \iff x = 0$ (positive definite);
2. $\forall \lambda \in \mathbb{R}$, $x \in X$,
$$\|\lambda x\| = |\lambda| \|x\| \qquad \text{(homogeneity)};$$
3. $\forall x, y \in X$,
$$\|x + y\| \leq \|x\| + \|y\| \qquad (\triangle\text{-inequality})$$

is called a norm on X. Any vector space with a norm is called a normed vector space.

Proposition

Every norm induces a metric via
$$d(x, y) := \|x - y\|.$$

Proof. Trivial. □

Example

1. $\forall p \in [1, \infty]$, $n \in \mathbb{N}$, $\|\cdot\|_p$ is a norm on $\mathbb{R}^n$.
2. Let $[a, b]$ be a compact interval. Let $C([a, b])$ be the continuous functions $[a, b] \to \mathbb{R}$. Define

the uniform norm

$$\|\cdot\|_\infty : f \longmapsto \max|f|.$$

(the extreme value theorem for continuous functions on a compact set)

Proof. For 2, consider the triangle inequality. Let $x \in [a,b]$.

$$|(f+g)(x)| \le |f(x)| + |g(x)| \le \|f\|_\infty + \|g\|_\infty$$

$$\implies \quad \|f+g\|_\infty \le \|f\|_\infty + \|g\|_\infty.$$

□

Lemma (Hölder's inequality)

$\forall p \in [1,\infty]$, $x, y \in \mathbb{R}^n$,

$$\sum_{i=1}^{n} x_i y_i \le \|x\|_p \|y\|_q,$$

where $q \in [1,\infty]$ is the Hölder conjugate:

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Proof. $p = \infty$:

$$\sum x_i y_i \le \sum |x_i y_i| \le \sum (\max|x_i|)|y_i| = \|x\|_\infty \|y\|_1.$$

$p = 1$ follows symmetric.

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$1 < p < \infty$: Assume $x \neq 0$, $y \neq 0$ (otherwise $0 = 0$). Let

$$x' = \frac{x}{\|x\|_p}, \qquad y' = \frac{y}{\|y\|_q}.$$

Note

$$\sum x_i y_i \le \|x\|_p \|y\|_q \quad \iff \quad \sum x_i' y_i' \le 1.$$

Young's Inequality

$$\forall a, b \in (0,\infty), \qquad ab \le \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof. First proof (weighted arithmetic–geometric mean inequality). Jensen's inequality:

$$A^t B^{1-t} \le tA + (1-t)B, \qquad t \in [0,1].$$

Define

$$f(x) = tx + (1-t)B - x^t B^{1-t}.$$

By differentiating it,

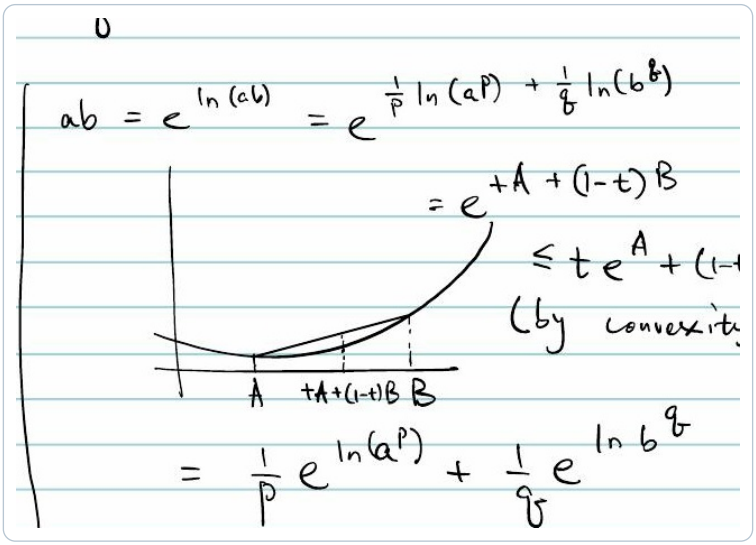
$$t - tx^{t-1}B^{1-t} = f'(x), \qquad f''(x) = t(1-t)x^{t-2}B^{1-t} \ge 0.$$

$\implies f$ attains min at $x = B$ by setting $f' = 0$. Then Jensen's follows.

Then note Young's $\sim$ follows by

$$A = a^p, \qquad B = b^q, \qquad t = \frac{1}{p}.$$

Second proof (convexity of the exponential function).



$$\begin{aligned} ab = e^{\ln(ab)} &= e^{\frac{1}{p}\ln(a^p)+\frac{1}{q}\ln(b^q)} = e^{tA+(1-t)B} \\ &\le te^A + (1-t)e^B \qquad \text{(by convexity)} \\ &= \frac{1}{p}e^{\ln(a^p)} + \frac{1}{q}e^{\ln(b^q)} = \frac{a^p}{p} + \frac{b^q}{q}. \end{aligned}$$

□

Thus

$$\begin{aligned} \sum x_i' y_i' \le \sum |x_i' y_i'| &\le \sum \left(\frac{|x_i'|^p}{p} + \frac{|y_i'|^q}{q} \right) \\ &= \sum \frac{|x_i|^p}{p\|x\|_p^p} + \sum \frac{|y_i|^q}{q\|y\|_q^q} = \frac{1}{p} + \frac{1}{q} = 1. \end{aligned}$$

□

Lemma (Minkowski Inequality)

$\forall p \in [1, \infty]$, $x, y \in \mathbb{R}^n$,

$$\|x + y\|_p \le \|x\|_p + \|y\|_p.$$

Proof. $p = \infty$ or 1 are trivial.

$\underline{1 < p < \infty}$:

$$\begin{aligned}\|x+y\|_p^p &= \sum |x_i + y_i|^p = \sum |x_i + y_i|^{p-1}|x_i + y_i| \\ &\le \sum |x_i + y_i|^{p-1}|x_i| + \sum |x_i + y_i|^{p-1}|y_i| \\ &\le \left(\sum (|x_i + y_i|^{p-1})^q\right)^{1/q} \left(\sum |x_i|^p\right)^{1/p} \\ &\quad + \left(\sum (|x_i + y_i|^{p-1})^q\right)^{1/q} \left(\sum |y_i|^p\right)^{1/p}\end{aligned}$$

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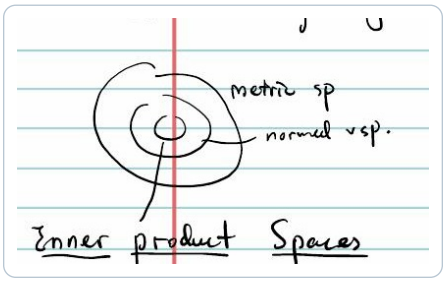
$$\begin{aligned}&= \left(\sum |x_i + y_i|^p\right)^{1/q} \|x\|_p + \left(\sum |x_i + y_i|^p\right)^{1/q} \|y\|_p \\ &= \|x+y\|_p^{p/q}(\|x\|_p + \|y\|_p). \\ \implies \quad &\|x\|_p + \|y\|_p \ge \|x+y\|_p^{p-p/q} = \|x+y\|_p^1.\end{aligned}$$

□

Proposition

$\forall 1 \le p \le \infty$, $\|\cdot\|_p$ is a norm on $\mathbb{R}^n$.

Proof. Everything else is trivial. The triangle inequality follows from Minkowski's inequality. □



2.3 Inner Product Spaces

Definition

Given a vector space X over $\mathbb{R}$, a function

$$\langle \cdot, \cdot \rangle : X \times X \longrightarrow \mathbb{R}$$

is an inner product on X if

(i) $\forall x \in X$,

$$\langle x, x \rangle \begin{cases} > 0, & \text{if } x \neq 0, \\ = 0, & \text{if } x = 0; \end{cases}$$

(ii) $\langle x, y\rangle = \langle y, x\rangle$;

(iii) $\forall x, y, z, \lambda_1, \lambda_2$,

$$\langle \lambda_1 x + \lambda_2 y, z\rangle = \lambda_1\langle x, z\rangle + \lambda_2\langle y, z\rangle.$$

Any vector space equipped with an inner product is called an inner product space. □

Remark. A Hermitian inner product on a complex vector space satisfies

$$\langle x, y\rangle = \overline{\langle y, x\rangle},$$

is linear in its first argument,

$$\langle \lambda_1 x + \lambda_2 y, z\rangle = \lambda_1\langle x, z\rangle + \lambda_2\langle y, z\rangle,$$

and is conjugate-linear in its second argument,

$$\langle x, \lambda_1 y + \lambda_2 z\rangle = \overline{\lambda_1}\langle x, y\rangle + \overline{\lambda_2}\langle x, z\rangle.$$

Remark.

$$\langle x, \lambda_1 y + \lambda_2 z\rangle = \lambda_1\langle x, y\rangle + \lambda_2\langle x, z\rangle$$

for a real vector space. □

Example

$$\langle x, y\rangle = \sum_{i=1}^{n} x_i y_i \quad \text{on } \mathbb{R}^n.$$

Example

On $\mathbb{R}^n$, given $A = (a_{ij})_{1\le i,j\le n}$, define

$$\langle x, y\rangle = \sum_{i,j=1}^{n} a_{ij} x_i y_j.$$

A is assumed to be symmetric and positive definite, which we define it as

$$\sum_{i,j=1}^{n} a_{ij} x_i x_j \ge 0, \quad \text{while } = 0 \iff x = 0$$

(We will leave the explanation

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on the simpler equivalent conditions to the algebraists.)

Symmetric matrices are always diagonalizable with $\mathbb{R}$-eigenvalues. A symmetric matrix is positive definite $\iff$ it has strictly positive eigenvalues.

□

Theorem (Cauchy–Schwarz)

Let $(X, \langle\cdot,\cdot\rangle)$ be a real inner product space. Let $x, y \in X$. Then

$$|\langle x, y\rangle| \leq \sqrt{\langle x, x\rangle\langle y, y\rangle}.$$

Proof. If $y = 0$, the inequality is immediate. Assume $y \neq 0$ and consider

$$f(t) = \langle x + ty, x + ty\rangle = \langle x, x\rangle + t^2\langle y, y\rangle + 2t\langle x, y\rangle.$$

f is smooth on $\mathbb{R}$. Differentiating it gives a minimum at

$$-\frac{\langle x, y\rangle}{\langle y, y\rangle}.$$

Substitution gives the inequality because $f(t) \geq 0$ by positive definiteness.

This proof extends to a Hermitian inner product space. For real t,

$$f(t) = \langle x, x\rangle + 2t\,\mathrm{Re}\langle x, y\rangle + t^2\langle y, y\rangle,$$

$$f\left(-\frac{\mathrm{Re}\langle x, y\rangle}{\langle y, y\rangle}\right) \geq 0 \quad \Longrightarrow \quad |\,\mathrm{Re}\langle x, y\rangle| \leq \sqrt{\langle x, x\rangle\langle y, y\rangle}.$$

If $\langle x, y\rangle = 0$, the result is immediate. Otherwise, let

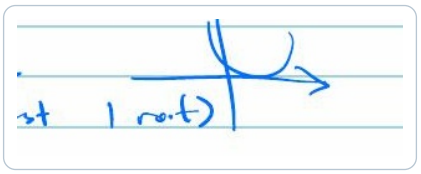
$$\lambda = \frac{\overline{\langle x, y\rangle}}{|\langle x, y\rangle|}, \qquad |\lambda| = 1.$$

Applying the preceding real-part inequality to λx and y gives

$$\begin{aligned} |\langle x, y\rangle| &= \lambda\langle x, y\rangle = \langle \lambda x, y\rangle \\ &\leq \sqrt{\langle \lambda x, \lambda x\rangle\langle y, y\rangle} = \sqrt{\lambda\overline{\lambda}\langle x, x\rangle\langle y, y\rangle} \\ &= \sqrt{1 \cdot \langle x, x\rangle\langle y, y\rangle}. \end{aligned}$$

□

Note. The real case can also be shown using the fact that the discriminant is at most zero, since a quadratic polynomial that is nonnegative everywhere has $\Delta \leq 0$.



Proposition

If $(X, \langle \cdot, \cdot \rangle)$ is a real inner product space, then $\| \cdot \| : X \to \mathbb{R}$ defined by

$$\|x\| := \sqrt{\langle x, x \rangle}$$

is a norm on X.

Proof. For the triangle inequality, note that

$$\begin{aligned} \|x+y\|^2 &= \langle x+y, x+y \rangle \\ &= \langle x, x \rangle + \langle y, y \rangle + 2\langle x, y \rangle \\ &\leq \langle x, x \rangle + \langle y, y \rangle + 2\sqrt{\langle x, x \rangle \langle y, y \rangle} \\ &= (\|x\| + \|y\|)^2. \end{aligned}$$

Other conditions are trivial. □

Definition

Let (X, d) be a general metric space, $x \in X$, and $\varepsilon \in (0, +\infty)$. We define the open ball centered at x, with radius ε, as

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$$B(x, \varepsilon) := \{y \in X \mid d(x, y) < \varepsilon\},$$

also denoted by $B_d(x, \varepsilon)$ or $B_{(X,d)}(x, \varepsilon)$.

Example

$X = \mathbb{R}^n$,

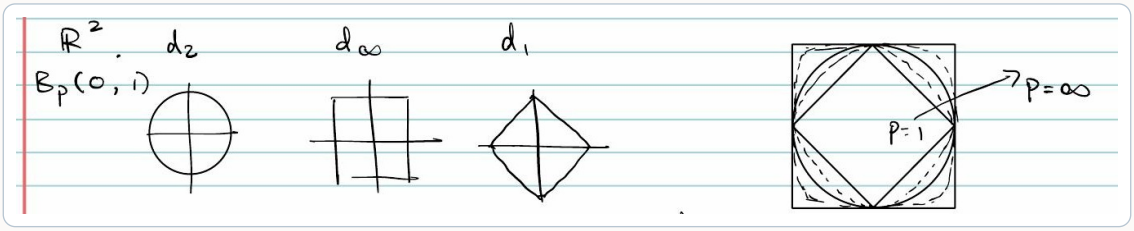
$$d_p(x, y) = \|x - y\|_p = \begin{cases} \left(\sum_{i=1}^{n} |x_i - y_i|^p \right)^{1/p}, & 1 \leq p < \infty, \\ \max\limits_{1 \leq i \leq n} |x_i - y_i|, & p = \infty. \end{cases}$$

d_2: Euclidean, d_∞: uniform, d_1: taxi cab / Manhattan. □

Remark. The higher p is, the smaller the impact of a small coordinate $|x_i|$ has to the norm $\|x\|_p$.

Example

$\mathbb{R}^2$, $B_p(0, 1)$:



Proven in Assignment 1, Question 5 that $\forall p, q$, $p < q$, we have

$$\|x\|_q \leq \|x\|_p, \qquad \forall x \in \mathbb{R}^n.$$

□

Example

$$d(x,y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y \end{cases} \qquad \text{(discrete metric)}$$

$$B_d(x,r) = \begin{cases} \{x\}, & r \leq 1, \\ X, & r > 1. \end{cases}$$

2.4 Open and Closed Sets

Definition

Let (X,d) be a metric space with $A \subseteq X$. We say A is open if

$$\forall x \in A,\ \exists r > 0 \text{ such that } B_d(x,r) \subseteq A.$$

We say A is closed if A^c is open. □

Remark. A set can be open and closed (clopen); a set can also be neither. □

Remark. Specify the metric. Different metrics induce different open sets, that is, different topologies. □

Proposition

Let (X,d) be a metric space, $x_0 \in X$, $r_0 \in (0,\infty)$, then

1. $B(x_0,r_0)$ is open in (X,d).
2. $$B[x_0,r_0] := \{x \in X \mid d(x_0,x) \leq r_0\}$$
   is closed in (X,d).

Proof. Trivial exercises using the triangle inequality and the definitions. □

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Remark. In $\mathbb{R}$, $d(x,y) := |x-y|$,

$$B_d\left(\frac{a+b}{2}, \frac{b-a}{2}\right) = (a,b).$$

Remark. For any set X,

$$d(x,y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

Every subset is open in (X,d). Consequently, all subsets are closed in (X,d).

Proposition

Let (X,d) be a metric space.

(a) X and ϕ are clopen.

(b) Given open sets $(A_i)_{i\in I} \subseteq (X,d)$, $\bigcup_{i\in I} A_i$ is open in (X,d).

(c) Given finite family of open sets $(A_i)_{i\in\{1,\dots,n\}} \subseteq (X,d)$, then $\bigcap_{i=1}^n A_i$ is open in (X,d).

(d) Given family of closed sets $(A_i)_{i\in I}$, $\bigcap_{i\in I} A_i$ is closed.

(e) Given finite $(A_i)_{i=1}^n$ closed, $\bigcup_{i=1}^n A_i$ is closed.

(f) A_1 open, A_2 closed in (X,d), then $A_1 \setminus A_2$ is open in (X,d).

(g) A_1 closed, A_2 open in (X,d), then $A_1 \setminus A_2$ is closed in (X,d).

Proof. Trivial. See Rudin's *Principles of Mathematical Analysis* if having difficulty. Hint:

$$A_1 \setminus A_2 = A_1 \cap A_2^c.$$

□

Counterexample. In $\mathbb{R}^n$,

$$\bigcap_{k=1}^{\infty} \underbrace{B_{d_p}\left(0, \frac{1}{k}\right)}_{\text{open}} = \{0\} \qquad \text{(why?)}.$$

In $\mathbb{R}^n$,

$$\bigcup_{k=1}^{\infty} \left(\mathbb{R}^n \setminus B_{d_p}\left(0, \frac{1}{k}\right)\right) = \mathbb{R}^n \setminus \{0\}$$

is not closed as its complement is not open. □

Definition

Let (X,d) be a metric space. Let $A \subseteq X$.

(a) We call the closure of A in (X,d) the set

$$\overline{A} := \{x \in X \mid B(x,r) \cap A \neq \phi,\ \forall r \in (0,\infty)\}.$$

(b) We call the interior of A in (X, d) the set

$$A^\circ := \{x \in A \mid \exists r \in (0, \infty) \text{ such that } B(x, r) \subseteq A\}.$$

(c) We call the boundary of A in (X, d) the set

$$\partial A := \overline{A} \setminus A^\circ.$$

□

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Remark. Equivalently,

$$A^\circ = (\overline{A^c})^c, \qquad \operatorname{Ext} A = (A^c)^\circ = (\overline{A})^c,$$
$$\partial A = (\operatorname{Ext} A \cup A^\circ)^c. \qquad \text{(verify directly from the definition)}$$

Example

If $(X, \|\cdot\|)$ is a normed vector space, $d(x, y) = \|x - y\|$,

$$\overline{B_d(x_0, r_0)} = \{x \in X \mid \|x - x_0\| \le r_0\},$$
$$\left(\overline{B_d(x_0, r_0)}\right)^\circ = B_d(x_0, r_0),$$
$$\partial B_d(x_0, r_0) = \{x \in X \mid \|x - x_0\| = r_0\}.$$

□

Discrete metrics.

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases} \quad \star$$

$$B(x_0, 1) = \{x_0\}, \qquad \overline{B_d(x_0, 1)} = \{x_0\}.$$

If $\exists x \in X \setminus \{x_0\}$ such that $\forall r \in (0, \infty)$,

$$B_d(x, r) \cap B_d(x_0, 1) \neq \phi,$$

then taking $r < 1$, $B_d(x, r) = \{x\} \implies \{x\} \cap \{x_0\} \neq \phi \implies x = x_0$, $\longrightarrow\!\longleftarrow$.

$$\{x \in X : d(x, x_0) \le 1\}^\circ = X^\circ = X, \quad \text{but NOT } B_d(x_0, 1)!$$

Unless X is a singleton itself, otherwise

$$\overline{B_d(x_0, 1)} \neq B_d[x_0, 1], \qquad B_d[x_0, 1]^\circ \neq B_d(x_0, 1)!$$

Lastly

$$\partial B_d(x_0, 1) = \{x_0\} \setminus \{x_0\} = \phi.$$

□

Proposition

Let (X, d) be a metric space, and $A \subseteq X$.

(i)
$$\overline{A^c} = (A^\circ)^c, \qquad (A^c)^\circ = (\overline{A})^c;$$

(ii)
$$\partial A = \overline{A} \cap \overline{A^c} = \partial(A^c).$$
$$\left[= X \setminus (\operatorname{Ext} A \cup A^\circ) = \underbrace{X \setminus \operatorname{Ext} A}_{=\overline{A}} \cap \underbrace{X \setminus A^\circ}_{=\overline{A^c}} \right]$$

Proof. (i) follows trivially from their definitions.

(ii):
$$\partial A = \overline{A} \setminus A^\circ = \overline{A} \cap (A^\circ)^c = \overline{A} \cap \overline{A^c}.$$

□

Proposition

Let (X, d) be a metric space, $A \subseteq X$. Then A° is the largest open set in (X, d) contained in A. That is,

1. $A^\circ \subseteq A$;
2. A° is open;
3. $\forall U \subseteq A$ open, $U \subseteq A^\circ$.

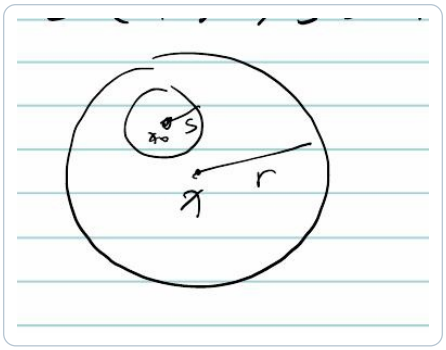
Moreover,
$$A = A^\circ \iff A \text{ is open in } (X, d).$$

Remark. The uniqueness follows from the properties themselves.

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Proof. 1 follows from the definition.

2 Let $x \in A^\circ$. Then $\exists r \in (0, \infty)$ such that $B(x, r) \subseteq A$. Since $B(x, r)$ is open, $\forall x_0 \in B(x, r)$, $\exists s \in (0, \infty)$ such that
$$B(x_0, s) \subseteq B(x, r) \subseteq A.$$
$$\implies x_0 \in A^\circ \implies B(x, r) \subseteq A^\circ.$$



3 Let $U \subseteq A$ be open. Let $x \in U$. Then

$$B(x,r) \subseteq U \subseteq A \Longrightarrow x \in A^\circ \Longrightarrow U \subseteq A^\circ.$$

Lastly, ($\Longleftarrow$) is direct since A itself is the largest open set in (X,d) containing itself, ($\Longrightarrow$) follows from the 2 property. □

Proposition

Let (X,d) be a metric space, $A \subseteq X$, the closure of A is the ③ smallest ② closed set in (X,d) which ① contains A.

Moreover

$$A = \overline{A} \Longleftrightarrow A \text{ is closed itself.}$$

Proof. Apply the previous two propositions to A^c, and use

$$(A^c)^\circ = (\overline{A})^c.$$

□

Corollary

Let (X,d) be a metric space, $A \subseteq X$, then

$$\partial A \text{ is closed in } (X,d).$$

Proof.

$$\partial A = \underbrace{\overline{A}}_{\text{closed}} \setminus \underbrace{A^\circ}_{\text{open}}.$$

□

Proposition

Let $(X, \|\cdot\|)$ be a normed vector space over $\mathbb{R}$ or $\mathbb{C}$. Then $\forall x_0, r_0$,

$$\overline{B_{\|\cdot\|}(x_0,r_0)} = B_{\|\cdot\|}[x_0,r_0],$$

$$\left(B_{\|\cdot\|}[x_0,r_0]\right)^\circ = B_{\|\cdot\|}(x_0,r_0),$$

$$\partial B_{\|\cdot\|}(x_0, r_0) = \{x \in X \mid \|x - x_0\| = r_0\}.$$

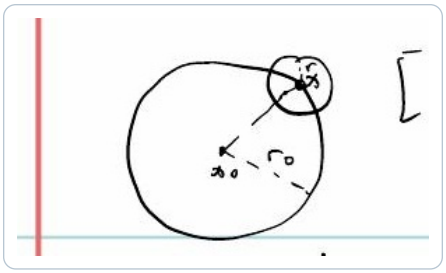
Proof. Closure is the smallest close set containing $B(x_0, r_0)$, $B[x_0, r_0]$ is closed containing $B(x_0, r_0)$

$$\implies \overline{B(x_0, r_0)} \subseteq B[x_0, r_0].$$

Let $x \in B[x_0, r_0]$. $\forall r > 0$, choose $\varepsilon > 0$ small enough such that

$$x_0 + (1 - \varepsilon)(x - x_0) \in B(x_0, r_0) \cap B(x, r).$$

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$$\left[\varepsilon = \frac{r}{2\|x - x_0\|}\right].$$

Thus by the definition,

$$B[x_0, r_0] \subseteq \overline{B(x_0, r_0)}.$$

Since interior is largest open set in $B[x_0, r_0]$, and $B(x_0, r_0) \subseteq B[x_0, r_0]$ is open, we have

$$B(x_0, r_0) \subseteq B[x_0, r_0]^\circ.$$

Let $x \in B[x_0, r_0]^\circ$. Assume

$$\|x - x_0\| = r_0 \qquad \text{(the only case where } x \notin B(x_0, r_0)).$$

$\forall r > 0$, choose $\varepsilon > 0$ small enough such that

$$x_\varepsilon = x_0 + (1 + \varepsilon)(x - x_0) \in B(x, r), \qquad x_\varepsilon \notin B[x_0, r_0].$$

Then

$$B(x, r) \nsubseteq B[x_0, r_0] \implies x \notin B[x_0, r_0]^\circ, \quad \longrightarrow\!\longleftarrow .$$

Thus

$$\|x - x_0\| < r_0 \implies x \in B(x_0, r_0).$$

$$\text{set is open} \iff A = A^\circ.$$

$$\begin{aligned}\partial B(x_0, r_0) &= \overline{B(x_0, r_0)} \setminus B(x_0, r_0)^\circ \\ &= B[x_0, r_0] \setminus B(x_0, r_0) \\ &= \{x \in X \mid \|x - x_0\| = r_0\}.\end{aligned}$$

□

3 Limits, Completeness, and Compactness

Definition

Let (X, d) be a general metric space, $x \in X$, and let $(x_k)_{k\in\mathbb{N}}$ be a sequence of elements in X. We say $(x_k)_k$ converges to x in (X, d), and write

$$x_k \xrightarrow{k\to\infty} x, \qquad \lim_{k\to\infty} x_k = x,$$

if

$$\forall \varepsilon > 0,\ \exists N_\varepsilon \in \mathbb{N} \text{ such that } \forall k \geq N_\varepsilon,\ d(x_k, x) < \varepsilon.$$

We say x the limit of $(x_k)_k$. □

Proposition

1. $$\lim_{k\to\infty} x_k = x \iff \lim_{k\to\infty} d(x_k, x) = 0.$$
   (in (X, d); in $\mathbb{R}$)
2. $\lim_{k\to\infty} x_k = x$ is unique.
3. $$\lim_{k\to\infty} x_k = x \iff \lim_{j\to\infty} x_{k_j} = x, \quad \forall \{x_{k_j}\}_j \subseteq \{x_k\}_k \quad (k_{j+1} > k_j,\ \forall j).$$

Proof. 1

$$\lim_{k\to\infty} x_k = x \iff \forall \varepsilon > 0,\ \exists N_\varepsilon \in \mathbb{N} \text{ such that } \forall k \geq N_\varepsilon,\ d(x_k, x) < \varepsilon$$
$$\iff \lim_{k\to\infty} d(x, x_k) = 0 \quad \text{(by the definition).} \qquad |d(x, x_k) - 0|$$

2 Assume $\exists x, x_0 \in X$ being limits of $\{x_k\}_k$.

$$d(x, x_0) \leq d(x, x_k) + d(x_0, x_k) \xrightarrow{k\to\infty} 0 + 0.$$

$$\implies d(x, x_0) = 0 \implies x = x_0 \quad \text{by positive definiteness.}$$

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3 ($\Longleftarrow$) Straightforward. $\{x_k\}$ is a subsequence of itself.

($\Longrightarrow$) Notice that

$$k_j \geq j, \qquad \forall j \in \mathbb{N},$$

given that $\{k_j\}_j \subseteq \mathbb{N}$ (induction). Then it follows from the definition. □

Example

$(\mathbb{R}^n, \|\cdot\|_p)$:

$$\lim_{k\to\infty} x_k = x \iff \text{coordinatewise convergence in } \mathbb{R}.$$

Note that

$$\lim_{k\to\infty} |x_i - (x_k)_i| \le \lim_{k\to\infty} \|x - x_k\|_p = 0$$

(the ith component; by the Squeeze Theorem). □

Example

(X, d_{discrete}). Then

$$\lim_{k\to\infty} x_k = x \iff \{x_k\} \text{ becomes constantly } x \text{ at the end.}$$

□

Example

Let I be a nonempty interval $\subseteq \mathbb{R}$. Let $\mathcal{B}(I)$ be the vector space of all bounded $\mathbb{R}$-valued functions, that is,

$$\sup_{x\in I} |f(x)| < \infty.$$

Equip $\mathcal{B}(I)$ with the uniform norm

$$\|f\|_\infty := \sup_{x\in I} |f(x)|.$$

Exercise: Show that $\|\cdot\|_\infty$ is indeed a norm on $\mathcal{B}(I)$.

Let $(f_k)_{k\in\mathbb{N}}$ be a sequence in $\mathcal{B}(I)$. Then

$$f_k \xrightarrow{k\to\infty} f \quad \text{in } (\mathcal{B}(I), \|\cdot\|_\infty)$$

$$\iff \forall \varepsilon > 0,\ \exists N_\varepsilon \in \mathbb{N} \text{ such that } \forall k \ge N_\varepsilon, \quad \|f_k - f\|_\infty < \varepsilon$$

$$\iff \lim_{k\to\infty} \|f_k - f\|_\infty = 0$$

$$\iff \forall \varepsilon > 0,\ \exists N_\varepsilon \in \mathbb{N} \text{ such that } \forall k \ge N_\varepsilon,\ \forall x \in I, \quad |f_k(x) - f(x)| < \varepsilon. \qquad (*)$$

□

Definition

We say f_k converges to f uniformly if $(*)$ holds. Equivalently, f_k converges to f in $(\mathcal{B}(I), \|\cdot\|_\infty)$. We call f the uniform limit of $\{f_k\}_{k=1}^\infty$. □

Remark 1: Compare this to pointwise convergence: $f_k \to f$ pointwise if

$$\forall x \in I,\ \forall \varepsilon > 0,\ \exists N_{\varepsilon,x} \in \mathbb{N} \text{ such that } \forall k \ge N_{\varepsilon,x},\ |f_k(x) - f(x)| < \varepsilon.$$

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2: We restricted $f_k, f \in \mathcal{B}(I)$ but that need NOT be the case. Consider

$$\underbrace{\sqrt{x^2+\frac{1}{n}}}_{f_n(x)} \quad \text{and} \quad \underbrace{|x|}_{f(x)} \quad \text{on } \mathbb{R},$$

both are unbounded yet uniform convergence occurs:

$$\left|\sqrt{x^2+\frac{1}{n}}-|x|\right| \leq \sqrt{\frac{1}{n}} \quad (\text{why?}) < \varepsilon$$

by choosing n large.

$$|\sqrt{a}-\sqrt{b}|^2 \leq |\sqrt{a}-\sqrt{b}|\,|\sqrt{a}+\sqrt{b}| = |a-b|.$$

□

Example

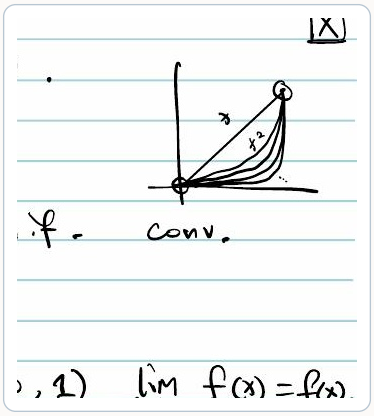
$I=(0,b)$, $0<b\leq 1$, $f_k(x)=x^k$, $f(x)=0$. Then

$$\|f_k - f\|_\infty = \sup|f_k(x)-f(x)| = b^k.$$

If $b<1$, then $\|f_k-f\|_\infty \to 0$. The convergence is uniform.

If $b=1$, then $\|f_k-f\|_\infty \to 1$. The convergence is not uniform, but it is pointwise:

$$\forall x \in (0,1), \qquad \lim_{k\to\infty} f_k(x) = f(x).$$



Proposition

Let (X,d) be a metric space, $x \in X$, $A \subseteq X$. Then

$$x \in \overline{A} \iff \exists \{x_k\}_{k=1}^{\infty} \subseteq A \text{ such that } \lim_{k\to\infty} x_k = x.$$

[Sequential characterisation for closure]

Proof. Exercise. This should be elementary from the definitions. □

Corollary

Let (X, d) be a metric space, $A \subseteq X$. Then the following are equivalent:

(1) A is closed in (X, d);

(2) If $\exists (x_k)_{k=1}^{\infty} \subseteq A$ such that $x_k \to x \in X$, then $x \in A$, that is, every limit point of A is a point in A.

Proof. Use the previous proposition and the fact that

$$A = \overline{A} \iff A \text{ is closed in } (X, d).$$

□

3.1 Completeness and Compactness

Definition

Let (X, d) be a metric space and $(x_k)_{k \in \mathbb{N}}$ a sequence in X. Then $(x_k)_k$ is Cauchy if

$$\forall \varepsilon > 0,\ \exists N_\varepsilon \in \mathbb{N} \text{ such that } \forall k, n \geq N_\varepsilon, \quad d(x_k, x_n) < \varepsilon.$$

Proposition

If $(x_k)_{k \in \mathbb{N}}$ converges in (X, d), then it is Cauchy in (X, d).

Proof. Exercise. Use $\triangle$-inequality. □

Definition

A metric space (X, d) is complete if every Cauchy sequence

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converges in (X, d).

A complete normed vector space is a Banach space.

A complete inner product space is a Hilbert space. □

Example

$\forall n \in \mathbb{N}$, $\forall p \in [1, \infty]$, $(\mathbb{R}^n, \|\cdot\|_p)$ is complete.

Proof. Exercise. Hint: Show that each component gives a Cauchy sequence in $\mathbb{R}$. Thus convergence in each component $\implies$ convergence in $(\mathbb{R}^n, \|\cdot\|_p)$. □ □

Example

Discrete metric (X, d). Then (X, d) is trivially complete. □

Remark. Notice the previous example illustrates the point that convergence and completeness, Cauchyness is relative to the metric. □

Example

$(\mathcal{B}(I), \|\cdot\|_\infty)$, $(C([a,b]), \|\cdot\|_\infty)$ is complete. □

Proposition

Let (X, d) be a metric space, $A \subseteq X$.

If (A, d) is complete, then A is closed in (X, d).

If (X, d) is complete, A is closed in (X, d), then (A, d) is complete.

Proof. (1) Assume that (A, d) is complete. Let $x \in X$ and $\{x_k\}_k$ be a sequence of elements in A such that $\lim x_k = x$. Then $\{x_k\}_k$ is Cauchy

$$\implies \lim_{k\to\infty} x_k = x \in A.$$

(2) Let $\{x_k\}_k \subseteq A$ be Cauchy. Then $x_k \to x \in X$. Yet x is a limit point of A, A is closed $\implies x \in A$. □

Definition

Let (X, d) be a metric space, $A \subseteq X$. We say A is sequentially compact in (X, d), or (A, d) is sequentially compact (compactness does not rely on the ambient space), if every sequence has a subsequence converging in (A, d).

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Proposition

Let (X, d) be a metric space, $A \subseteq X$ compact, $\{x_k\}_k$ be a sequence in A and $x \in A$. Then

$$\lim_{k\to\infty} x_k = x \iff \text{Every convergent subsequence of } \{x_k\}_k \text{ converges to } x.$$

Proof. ($\implies$) is trivial.

($\impliedby$) Suppose x_k does not converge to x. Then $\exists \varepsilon_0 > 0$ such that

$$\forall n \in \mathbb{N}, \quad \exists k_n \geq n \text{ such that } d(x_{k_n}, x) > \varepsilon_0.$$

Then we have a sequence $\{x_{k_n}\}_{n=1}^\infty$ that is always bounded away from x by ε_0. Yet since A is compact, $\exists \{x_{k_{n_j}}\}_{j=1}^\infty$ that converges in (A, d)

$$\implies x_{k_{n_j}} \xrightarrow{j\to\infty} x$$

by our assumption. $\longrightarrow\longleftarrow$. Thus $\lim_{k\to\infty} x_k = x$. □

Remark. Note that, in general, a sequence may not even have any convergent subsequence. Compactness supplies this missing condition. □

Proposition

Let (X, d) be a metric space. Let $A \subseteq X$ be compact.

(1) A is closed in (X, d).

(2) A is bounded in (X, d), that is, $\exists C > 0$ such that $\forall x, y \in A$,

$$d(x, y) \leq C.$$

Equivalently,

$$\sup_{x,y} d(x, y) < \infty.$$

(3) (A, d) is complete.

Proof. (1) If $x \in X$, $\{x_k\} \subseteq A$, $x_k \to x$, then $\exists x_{k_j} \to x \in A$ by compactness and uniqueness of limit.

(2) Suppose A is unbounded. First notice that A is bounded $\iff \forall x \in A$,

$$\exists C_x > 0 \text{ such that } \forall y \in A, \qquad d(x, y) < C_x.$$

An alternative trick:

$$\begin{array}{c} \forall k \in \mathbb{N},\ \exists\, d(x_k, y_k) > k \\ \downarrow \text{ compact} \\ x_{k_j} \longrightarrow y_{k_j} \\ \downarrow \text{ compact} \\ x_{k_{j_\ell}} \longleftarrow y_{k_{j_\ell}} \\ \underbrace{\qquad\qquad}_{\text{now both converge.}} \end{array}$$

Suppose $\exists x \in A$ such that $\forall n \in \mathbb{N}$, $\exists y_n \in A$ such that

$$d(x, y_n) > n.$$

Then $\exists \{y_{n_k}\}_{k=1}^{\infty}$ a convergent subsequence of y_n by compactness. Yet then $\{d(x, y_{n_k})\} > \{n_k\}_{k=1}^{\infty}$ becomes bounded. $\longrightarrow\longleftarrow$.

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(3) Let $\{x_n\}_n$ be Cauchy. By compactness $\exists \{x_{n_k}\}_k$ that converges in (A, d). Since every Cauchy sequence converges to the same limit as its subsequences, x_n converges to the same limit. □

I just realized the proof did not cover what we used in 3:

Lemma

If $\{x_k\}_k$ is Cauchy in (X,d), and $\exists\{x_{k_j}\}_j \to x \in X$, then

$$x_k \xrightarrow{k\to\infty} x.$$

Proof. Let $\varepsilon > 0$. Since $\{x_k\}$ is Cauchy, there exists $N \in \mathbb{N}$ such that

$$d(x_k, x_m) < \frac{\varepsilon}{2}, \qquad k, m \geq N.$$

Since $x_{k_j} \to x$, choose j such that $k_j \geq N$ and $d(x_{k_j}, x) < \varepsilon/2$. Then, for every $k \geq N$,

$$d(x_k, x) \leq \underbrace{d(x_k, x_{k_j})}_{<\varepsilon/2} + \underbrace{d(x_{k_j}, x)}_{<\varepsilon/2} < \varepsilon.$$

□

Proposition

Let (X,d) be a compact metric space. Then

$$A \subseteq X \text{ is closed} \iff A \text{ is compact in } (X,d).$$

Proof. We have shown $\Longleftarrow$ last time.

($\Longrightarrow$) Suppose A is closed. Every sequence $(x_n)_n \subseteq A$ has a subsequence $(x_{n_k})_k$ that converges in X, by compactness of X. Because A is closed, the limit of this subsequence belongs to A. Thus every sequence in A has a convergent subsequence in A, $\Longrightarrow A$ is compact. □

Theorem (Heine–Borel Theorem)

Let $(V, \|\cdot\|)$ be a finite-dimensional normed vector space. Then

$$A \subseteq V \text{ is compact} \iff A \text{ is closed and bounded in } (V, \|\cdot\|).$$

Remark. It is never true in an infinite-dimensional normed vector space.

(It requires extra conditions, for example, uniformly small tails with respect to an orthonormal basis, finite-dimensional subspace approximability, or uniform equicontinuity.)

Proof. A compact $\Longrightarrow A$ closed and bounded is already proven.

($\Longleftarrow$) Let $\{x_n\}_n$ be a sequence in A. Let $\{e_1, \ldots, e_n\}$ be a Hamel basis of V and $(x_{k,1}, \ldots, x_{k,n})$ be the coordinates of x_k with respect to this basis.

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Since $x_k \in A$, $\|x_k\| \leq C$ for every k, by boundedness.

Consider the simpler case in $(\mathbb{R}^n, \|\cdot\|_p)$ with the standard basis. Then

$$|x_i| \leq \|x\|_p, \qquad \forall x, i.$$

Therefore, the sequences of coordinates are all bounded in $\mathbb{R}$. By the Bolzano–Weierstrass theorem, there exists a subsequence of $\{x_{k,1}\}$ that converges in $\mathbb{R}$. Applying the Bolzano–Weierstrass theorem again, there exists a subsequence of the subsequence just obtained for which the corresponding second-coordinate sequence converges. Applying the Bolzano–Weierstrass theorem inductively to all n coordinates, we obtain a subsequence $\left\{x_{k_j}\right\}$ such that

$$\left\{x_{k_j,i}\right\} \text{ converges in } \mathbb{R} \text{ for every } i \in \{1, \ldots, n\}.$$

Thus $\left\{x_{k_j}\right\}$ converges in $\mathbb{R}^n$. Since A is closed, it follows that its limit $\in A \Rightarrow A$ is compact.

General Case.

Claim: $m_k = \max_{1\leq i\leq n} |x_{k,i}| \leq B$ for every k.

Suppose, for a contradiction, that (m_k) is unbounded. Then there exists a subsequence (m_{k_j}) such that

$$m_{k_j} := \max_{1\leq i\leq n} \left|x_{k_j,i}\right| \longrightarrow \infty \qquad \text{as } j \longrightarrow \infty.$$

Define

$$y_{k_j} := \left(\frac{x_{k_j,1}}{m_{k_j}}, \ldots, \frac{x_{k_j,n}}{m_{k_j}}\right) \in \mathbb{R}^n.$$

The sequence (y_{k_j}) is bounded in $\mathbb{R}^n$. By what we have proved in $\mathbb{R}^n$, after passing to a further subsequence and relabelling it, there exists $y \in \mathbb{R}^n$ such that

$$\lim_{j\to\infty} y_{k_j} = y.$$

Then

$$\left\|\sum_{i=1}^n y_i e_i\right\| \leq \left\|\sum_{i=1}^n (y_i - y_{k_j,i}) e_i\right\| + \left\|\sum_{i=1}^n \underbrace{\frac{x_{k_j,i}}{m_{k_j}}}_{y_{k_j,i}} e_i\right\|.$$

The first term tends to 0 as $j \to \infty$, while

$$\frac{\left\|x_{k_j}\right\|}{m_{k_j}} \leq \frac{C}{m_{k_j}} \downarrow 0 \qquad (j \to \infty).$$

Thus

$$\left\|\sum_{i=1}^n y_i e_i\right\| = \|y\| = 0 \Rightarrow y_i = 0, \quad \forall i = 1, 2, \ldots, n.$$

On the other hand,

$$\max_{1\leq i\leq n} \left|y_{k_j,i}\right| = \frac{\max_{1\leq i\leq n} \left|x_{k_j,i}\right|}{m_{k_j}} = \frac{m_{k_j}}{m_{k_j}} = 1.$$

Since $z \mapsto \max_{1\le i\le n} |z_i|$ is continuous, taking the limit gives

$$\max_{1\le i\le n} |y_i| = 1,$$

which contradicts $y_i = 0$ for every i.

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Thus

$$m_k = \max_{1\le i\le n} |x_{k,i}| \le C, \qquad \text{for some } C \ge 0.$$

Then, by the $\mathbb{R}^n$ case again, there exists a subsequence $\left\{x_{k_j}\right\}_j$ of $\{x_k\}_k$ such that

$$\lim_{j\to\infty} x_{k_j,i} = x_i, \quad \text{for every } i, \qquad \text{for some } x_i \in \mathbb{R}.$$

Let $x = \sum_{i=1}^n x_i e_i$. Then

$$\begin{aligned} \left\|x_{k_j} - x\right\| &\le \left\|\sum_{i=1}^n (x_{k_j,i} - x_i)e_i\right\| \\ &\le \sum_{i=1}^n \left|x_{k_j,i} - x_i\right| \|e_i\| \longrightarrow 0, \end{aligned}$$

where the last expression tends to zero because it is a finite sum. Therefore

$$x_{k_j} \xrightarrow{j\to\infty} x \quad \text{in } (V, \|\cdot\|).$$

It follows since A is closed. □

Theorem

Let (X, d) be a metric space and $A \subseteq X$. Then

$$A \text{ is sequentially compact} \iff \text{Every open cover } \{U_i\}_{i\in I} \text{ of } A, \ \exists \text{ a finite subcover}$$

(that is, a finite open cover that covers A).

Open cover:

$$A \subseteq \bigcup_{i\in I} U_i.$$

Proof. ($\Leftarrow$) Let $\{x_k\}_{k=1}^\infty$ be a sequence in A with no convergent subsequence. Then, for every $x \in X$, there exists $B_d(x, \varepsilon(x))$ that contains finitely many points of $\{x_k\}_k$. Now take

$$\bigcup_{x\in A} B_d(x, \varepsilon(x)),$$

then it is an open cover of $A \Rightarrow \exists$ finite subcover, yet $\{x_k\}$ is contained by this finite collection where each open set contains finitely many elements of $\{x_k\}$, which would imply that $\{x_k\}$ is finite $\to\leftarrow$. Thus A has to be sequentially compact.

($\Rightarrow$) Suppose A is compact and $\{U_i\}_{i\in I}$ is an open cover of A.

Claim: There exists $r > 0$ such that for every $x \in A$, there exists $i \in I$ such that $B(x, r) \subseteq U_i$.

($*$) Suppose otherwise. For every $k \in \mathbb{N}$, there exists $x_k \in A$ such that, for every $i \in I$,

$$B\left(x_k, \frac{1}{k}\right) \nsubseteq U_i.$$

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Since A is sequentially compact, there exist $x \in A$ and a subsequence $\left\{x_{k_j}\right\}_j$ of $\{x_k\}$ such that

$$\lim_{j\to\infty} x_{k_j} = x \in A.$$

Since $\{U_i\}$ covers $A \Rightarrow x \in U_i$ for some $i \in I$. By $(*)$, for every $j \in \mathbb{N}$ and every $i \in I$,

$$B\left(x_{k_j}, \frac{1}{k_j}\right) \nsubseteq U_i.$$

Yet this fails as they are all going to get close to x and sit inside the U_i containing x. $\rightarrow\leftarrow$.

Choose $x_1 \in A$. If $A \nsubseteq B(x_1, r)$, choose $x_2 \in A \setminus B(x_1, r)$ (r from last claim).

By induction continue choosing until A is covered.

We claim A must be able to be covered by finitely many of these. Otherwise, we can construct a sequence $\{x_k\}_{k=1}^{\infty}$ such that

$$A \nsubseteq \bigcup_{i=1}^{k} B(x_i, r), \qquad \forall k \in \mathbb{N}.$$

Since A is compact, there exists a subsequence $\left\{x_{k_j}\right\}_j$ such that

$$\lim_{j\to\infty} x_{k_j} = x \in A.$$

By how we chose x_k inductively,

$$x_{k+1} \notin \bigcup_{i=1}^{k} B(x_i, r) \Rightarrow d(x_k, x_n) \geq r, \quad \text{whenever } k \neq n,$$

$$\Rightarrow \left\{x_{k_j}\right\}_j \text{ can not converge as it is not Cauchy } \rightarrow\leftarrow .$$

Thus A has a finite subcover. □

Corollary

Let $A \subseteq (X, d)$. If A is compact, then for every $r > 0$ there exist N_r elements $x_1, \ldots, x_{N_r}$ in A such that

$$A \subseteq \bigcup_{i=1}^{N_r} B(x_i, r).$$

Proof. Note that $A \subseteq \bigcup_{x\in A} B(x, r)$ is an open cover. □

Proposition

Let $(V, \|\cdot\|)$ be an infinite-dimensional normed vector space. Then $\overline{B(0,1)}$ is not compact.

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Proof. Suppose it is compact. Then there exist $N \in \mathbb{N}$ and $\{x_i\}_{i=1}^N \subseteq \overline{B(0,1)}$ such that

$$\overline{B(0,1)} \subseteq \bigcup_{i=1}^{N} B\left(x_i, \frac{1}{2}\right).$$

Since $\dim(V) = \infty$, $\exists x_0 \notin \operatorname{span}\{x_i\}_{i=1}^N$. By multiplying x_0 by a nonzero scalar if necessary, assume without loss of generality that $0 < \|x_0\| < \frac{1}{2}$. Then $2x_0 \in \overline{B(0,1)}$ by homogeneity. Thus there exists $i \in \{1, 2, \ldots, N\}$ such that

$$\|2x_0 - x_i\| < \frac{1}{2} \Rightarrow \left\|x_0 - \frac{x_i}{2}\right\| < \frac{1}{4}, \quad \text{that is, } x_0 = \frac{x_i}{2} + y_1,\ \|y_1\| < \frac{1}{4}.$$

By induction, we can construct $z_k \in \operatorname{span}\{x_1, \ldots, x_N\}$ and $y_k \in V$ such that

$$x_0 = z_k + y_k \qquad \text{and} \qquad \|y_k\| < 2^{-k-1}.$$

(Then,

$$\left\|2^{k+1} y_k\right\| < 1 \Rightarrow 2^{k+1} y_k \in \overline{B(0,1)} \Rightarrow \left\|2^{k+1} y_k - x_j\right\| < \frac{1}{2}$$

$$\Rightarrow y_k = \frac{x_j}{2^{k+1}} + y_{k+1}, \quad \text{then back substitute, } z_{k+1} = z_k + \frac{x_j}{2^{k+1}}.)$$

It follows that

$$0 \leq \|z_k - x_0\| = \|y_k\| < \frac{1}{2^{k+1}} \longrightarrow 0,$$

and hence $z_k \to x_0$. Since $z_k \in \operatorname{span}\{x_1, \ldots, x_N\}$,

$$x_0 \in \overline{\operatorname{span}\{x_1, \ldots, x_N\}}.$$

We lastly claim: Every finite-dimensional normed vector space over a complete field is a Banach space. Let $(V, \|\cdot\|)$ be a finite-dimensional real or complex normed vector space. Let $\{a_k\}_{k=1}^\infty$ be a Cauchy sequence. It is bounded, so for some $r > 0$,

$$\{a_k\}_{k=1}^\infty \subseteq B(0,r) \subseteq \overline{B(0,r)}.$$

By the finite-dimensional Heine–Borel theorem proved above, $\overline{B(0,r)}$ is compact. Hence (a_k) has a convergent subsequence; because (a_k) is Cauchy, the entire sequence converges. Therefore $(V, \|\cdot\|)$ is complete. Since $\operatorname{span}\{x_1, \ldots, x_N\}$ is finite-dimensional over a complete field, it is complete and hence closed. Therefore

$$x_0 \in \overline{\operatorname{span}\{x_1, \ldots, x_N\}} = \operatorname{span}\{x_1, \ldots, x_N\},$$

a contradiction. □

A proof of this is done using homeomorphism from $V \to \mathbb{R}^n$ in 18.102.

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4 Continuity

Definition

Let (X, d) be a metric space, $x_0 \in X$, and $A \subseteq X$. We say x_0 is a cluster / limit point of A if

$$x_0 \in \overline{A \setminus \{x_0\}}.$$

Proposition

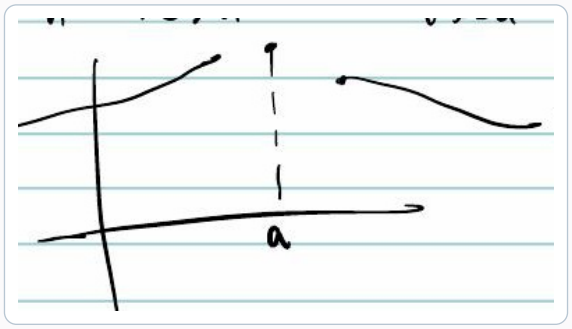
x_0 is a cluster point of A

$$\iff \exists \text{ a sequence } \{x_n\}_{n=1}^{\infty} \subseteq A \setminus \{x_0\}, \qquad x_n \xrightarrow{n\to\infty} x_0.$$

Proof. By the sequential characterization of closures. □

Remark

For example, in $\mathbb{R}$, the reason we used this definition is to avoid the case of isolated point:



f is defined at the point a, an isolated point.

Example

No point in a metric space equipped with the discrete metric is a cluster point.

Definition

Let (X, d_X) and (Y, d_Y) be two metric spaces, $A \subseteq X$, let x_0 be a cluster point of A, and let $y_0 \in Y$.

(1) For $f : A \to Y$, we say that $f(x)$ converges to y_0 as $x \to x_0$ and write

$$\lim_{x\to x_0} f(x) = y_0 \qquad \text{or} \qquad f(x) \longrightarrow y_0 \text{ as } x \to x_0$$

if

$$\forall \varepsilon > 0,\ \exists \delta > 0 \text{ such that } \forall x \in A, \quad 0 < d_X(x, x_0) < \delta, \quad d_Y(f(x), y_0) < \varepsilon.$$

(2) If $(Y, d_Y) = (\mathbb{R}, |\cdot|)$, for $f : A \to \mathbb{R}$, we say that $f(x)$ tends to $+\infty$ $(-\infty)$ as $x \to x_0$, write

$$\lim_{x \to x_0} f(x) = +\infty \ (-\infty) \qquad \text{or} \qquad f(x) \to +\infty \ (-\infty)$$

as $x \to x_0$ if $\forall C > 0$, $\exists \delta > 0$ such that $\forall x \in A$,

$$0 < d_X(x, x_0) < \delta \Rightarrow f(x) > C \quad (< -C).$$

(3) When $(X, d_X) = (\mathbb{R}, |\cdot|)$ and (c, ∞) (or $(-\infty, c)$) $\subseteq A$, for some $c \in \mathbb{R}$. We say f tends to y_0 as $x \to +\infty$ (or $-\infty$) if $\forall \varepsilon > 0$, $\exists C > 0$ such that $\forall x \in A$, $x > C$ (or $< -C$), we have

$$d_Y(f(x), y_0) < \varepsilon.$$

Proposition

Limit of a function is unique.

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Proof. Use contradiction and $\triangle$-inequality. $\square$

Definition

Let (X, d_X) and (Y, d_Y) be two metric spaces, $A \subseteq X$ and $x_0 \in A$. We say a function $f : A \to Y$ is continuous at x_0 if

$$\forall \varepsilon > 0,\ \exists \delta > 0 \text{ such that } \forall x \in A, \quad d(x, x_0) < \delta \Rightarrow d_Y(f(x), f(x_0)) < \varepsilon,$$

that is,

$$f(A \cap B_X(x_0, \delta)) \subseteq B_Y(f(x_0), \varepsilon).$$

If $B \subseteq A$, f is continuous on B if f is continuous at every point $x \in B$. If $B = A$, we just say f is continuous.

Proposition

If $x_0 \in A$ is a cluster point, then

$$\lim_{x \to x_0} f(x) = f(x_0) \iff f \text{ is continuous at } x_0.$$

Example

For every metric space (X, d), the identity function is continuous. (Exercise)

Example

$d(\,\cdot\,, y)$ and $d(x,\,\cdot\,)$ where d is a metric on the metric space (X, d) (some $x, y \in X$).

$$\forall x, x_0, \qquad d(x, y_0) \le d(x, x_0) + d(x_0, y_0),$$

$$d(x_0, y_0) \le d(x, x_0) + d(x, y_0),$$

$$|d(x, y_0) - d(x_0, y_0)| \le d(x, x_0) \Rightarrow \text{Lipschitz}.$$

Example

For every normed vector space $(V, \|\cdot\|)$, the norm $\|\cdot\|$ is continuous on $(V, \|\cdot\|)$.

Exercise: use the triangle inequality.

Example

Every inner-product space $(V, \langle\cdot,\cdot\rangle)$ has $\langle\cdot,\cdot\rangle$ continuous in each argument. Exercise: use the Cauchy–Schwarz inequality.

Example

Let (X, d) be a metric space, $A \subseteq X$, $x_0 \in A$, $n \in \mathbb{N}$, $p \in [1, \infty]$. Then

$$f : A \to (\mathbb{R}^n, \|\cdot\|_p)$$

is continuous $\iff f = (f_1, f_2, \ldots, f_n)$, each component $A \to \mathbb{R}$ is continuous. Exercise.

Proposition

Let (X, d_X), (Y, d_Y), (Z, d_Z) be metric spaces, $A \subseteq X$, $x_0 \in A$, $f : A \to Y$, $g : Y \to Z$.

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If f is continuous at x_0 and g is continuous at $f(x_0)$, then $g \circ f$ is continuous at x_0.

If f is continuous on B and g is continuous on $f(B)$, then $g \circ f$ is continuous on B.

Proof. This follows by applying the definition twice. □

Theorem

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$. Then

$$f \text{ is continuous} \iff \text{Preimage of open sets in } Y \text{ is open in } X$$

$$\iff \text{Preimage of closed sets in } Y \text{ is closed in } X.$$

Remark

A open $\not\Rightarrow f(A)$ is open (such maps are called open mappings).

For example, $f = \dfrac{1}{1+x^2}$ contradicts both.

Proof. ($\Rightarrow$) Suppose f is continuous. Let $A \subseteq Y$ be open. Let $x \in f^{-1}(A)$, $y = f(x)$. Then $\exists B_{d_Y}(y, \varepsilon) \subseteq A$. By continuity, there exists $B_{d_X}(f^{-1}(y), \delta)$ such that

$$f\left(B_{d_X}(f^{-1}(y), \delta)\right) \subseteq B_{d_Y}(y, \varepsilon)$$

$$\Rightarrow B_{d_X}(f^{-1}(y), \delta) \subseteq f^{-1}(B_{d_Y}(y, \varepsilon)) \subseteq f^{-1}(A)$$

$$\Rightarrow f^{-1}(A) \text{ is open.}$$

($\Leftarrow$) Let $f^{-1}(A)$ be open for A open. Then let $x \in X$, take open ball $B_{d_Y}(f(x), \varepsilon) \subseteq Y$. Then

$$x \in f^{-1}\left(B_{d_Y}(f(x), \varepsilon)\right) \text{ is open} \Rightarrow \exists B_{d_X}(x, \delta) \text{ that is} \subseteq f^{-1}\left(B_{d_Y}(f(x), \varepsilon)\right)$$

$$\Rightarrow f(B_{d_X}(x, \delta)) \subseteq B_{d_Y}(f(x), \varepsilon) \Rightarrow \text{continuity.}$$

Lastly, A open $\iff A^c$ closed, and $f^{-1}(A^c) = (f^{-1}(A))^c$. □

Proposition (Sequential characterization of continuity)

Let (X, d_X) and (Y, d_Y) be metric spaces, $A \subseteq X$, $x_0 \in A$, and $f : A \to Y$.

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Then f is continuous at $x_0 \iff \forall \{x_k\}_k \subseteq A$, $x_k \to x$, we have

$$\lim_{k\to\infty} f(x_k) = f(x).$$

Proof. Exercise. For ($\Rightarrow$), give a direct proof. For ($\Leftarrow$), use a proof by contradiction.

□

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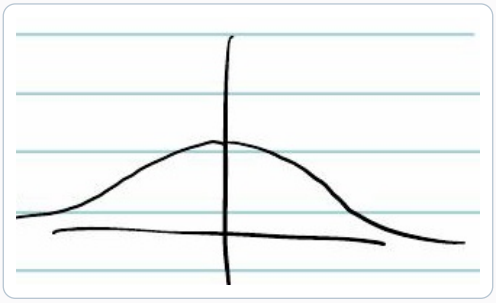
Theorem

Let (X, d_X) and (Y, d_Y) be metric spaces, let A be a compact set in (X, d_X), and let $f : X \to Y$ be continuous on A. Then $f(A)$ is compact.

Remark

Converse is not true.

$f(A)$ is compact $\not\Rightarrow$ A compact. For example, $f(x) = \dfrac{1}{1+x^2}$.



Proof. Direct consequence from sequential compactness and sequential characterization of continuity. □

Theorem

Let (X, d) be a metric space, and let $\emptyset \neq A \subseteq X$ be compact. If $f : A \to (\mathbb{R}, |\cdot|)$ is continuous on A, then it attains its maximum and minimum on A.

Proof. By the previous result, $f(A) \subseteq \mathbb{R}$ is compact. Since it is bounded and nonempty because $A \neq \emptyset$, the least-upper-bound property of $\mathbb{R}$ implies that $\sup f(A)$ and $\inf f(A)$ exists in $\mathbb{R}$.

By the properties of the infimum and supremum, there exist sequences $\{x_k\}_k$ and $\{y_k\}$ converging to them, respectively. Since $f(A)$ is closed,

$$\lim_{k\to\infty} x_k = \sup f(A) \in f(A),$$

$$\lim_{k\to\infty} y_k = \inf f(A) \in f(A).$$

$\Rightarrow$ the maximum and minimum are attained. □

Remark

We can formally define

$$\|f\|_\infty = \max_A |f|, \qquad A \text{ compact.}$$

And then $f(A)$ forms a Banach space.

Definition

Let (X, d_X) and (Y, d_Y) be metric spaces, $A \subseteq X$, and $f : A \to Y$. We

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4.1 Uniform Continuity

say f is uniformly continuous on A if

$$\forall \varepsilon > 0,\ \exists \delta > 0 \text{ such that } \forall x, y \in A, \quad d_X(x,y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon.$$

We say f is Lipschitz on A if there exists $C > 0$ such that

$$\forall x, y \in A, \qquad d_Y(f(x), f(y)) \leq C d_X(x,y).$$

Remark

Lipschitz $\Rightarrow$ uniformly continuous $\Rightarrow$ continuous.

Example

In the section on examples of continuity, we saw that every distance function is Lipschitz. Same with inner products.

Theorem (Heine–Cantor)

Let (X, d_X) and (Y, d_Y) be metric spaces, let $A \subseteq X$ be compact, and let $f : A \to Y$. Then f is continuous on A if and only if f is uniformly continuous on A.

Proof. ($\Leftarrow$) is trivial.

($\Rightarrow$) Suppose, for a contradiction, that f is not uniformly continuous. Then there exists $\varepsilon_0 > 0$ such that, for every $n \in \mathbb{N}$ ($\delta = \frac{1}{n}$), there exist $x_n, y_n \in A$ such that

$$(*) \qquad d_X(x_n, y_n) < \frac{1}{n}, \qquad d_Y(f(x_n), f(y_n)) \geq \varepsilon_0.$$

By compactness of A, there exists $\{x_{n_k}\}_k \subseteq \{x_n\}$ such that $x_{n_k} \to x \in A$.

Similarly, there exists $\left\{y_{n_{k_j}}\right\}_j \subseteq \{y_{n_k}\}_k \subseteq \{y_n\}_n$ such that

$$y_{n_{k_j}} \longrightarrow y.$$

Yet

$$d_X(x_{n_{k_j}}, y_{n_{k_j}}) < \frac{1}{n_{k_j}} \Rightarrow x = y.$$

By continuity of f, for every $\eta > 0$ there exists $N \in \mathbb{N}$ such that, for every $j \geq N$,

$$\begin{aligned} d_Y(f(x_{n_{k_j}}), f(y_{n_{k_j}})) &\leq d_Y(f(x_{n_{k_j}}), f(x)) + d_Y(f(y), f(y_{n_{k_j}})) \\ &< \eta. \end{aligned}$$

Thus a contradiction to $(*)$. □

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4.2 Contraction Mappings

Definition

Let (X, d_X) and (Y, d_Y) be metric spaces, $A \subseteq X$, and $f : A \to Y$. We say f is contractive / contraction (mapping) if there exists $C \in (0,1)$ such that

$$\forall x, y \in A, \qquad d_Y(f(x), f(y)) \leq C d_X(x, y).$$

Remark

Contractive mapping $\Rightarrow$ Lipschitz.

Theorem (Contraction Mapping Theorem / Banach Fixed-Point Theorem)

Let (X, d) be a nonempty complete metric space, and let $f : X \to X$ be a contractive mapping on X. Then there exists a unique $x_0 \in X$ such that $f(x_0) = x_0$.

Proof. Let $x_1 \in X$. Define recursively $f(x_n) =: x_{n+1}$. We claim $\{x_n\}_{n=1}^{\infty}$ is Cauchy. Let $\varepsilon > 0$. Suppose that f has contraction constant C.

Choose $N \in \mathbb{N}$ such that

$$N \geq \log_C \left(\frac{1-C}{d(x_1, x_2)} \varepsilon \right) + 1.$$

Let $n \geq m \geq N$. Then

$$\begin{aligned}
d(x_{n+1}, x_{m+1}) &= d(f(x_n), f(x_m)) \\
&\leq d(f(x_n), f(x_{n-1})) + \cdots + d(f(x_{m+1}), f(x_m)) \\
&\leq C d(x_n, x_{n-1}) + C d(x_{n-1}, x_{n-2}) + \cdots + C d(x_m, x_{m-1}) \\
&= C d(f(x_{n-1}), f(x_{n-2})) + C d(f(x_{n-2}), f(x_{n-3})) + \cdots + C d(f(x_{m-1}), f(x_{m-2})) \\
&\leq \cdots \qquad \text{(formally, this would be a proof by induction)} \\
&\leq C^n d(x_1, x_2) + C^{n-1} d(x_1, x_2) + \cdots + C^{m-1} d(x_1, x_2) \\
&= d(x_1, x_2)\, C^{m-1}(1 + C + \cdots + C^{n-m+1}) \\
&\leq d(x_1, x_2)\, C^{N-1} \left(\sum_{i=0}^{\infty} C^i \right) \qquad [m \geq N,\ C \in (0,1)] \\
&\leq d(x_1, x_2) \cdot \frac{1-C}{d(x_1, x_2)} \varepsilon \cdot \frac{1}{1-C} = \varepsilon.
\end{aligned}$$

Therefore indeed $\{x_n\}_{n=1}^{\infty}$ is Cauchy. By completeness, there exists $x_0 \in X$ such that $x_n \to x_0$. Yet by continuity,

$$\lim_{n\to\infty} f(x_{n-1}) = f\left(\lim_{n\to\infty} x_{n-1} \right) = f(x_0), \qquad \lim x_n = x_0.$$

Uniqueness: Suppose there exists $x' \in X$ such that $x' \neq x_0$ and $f(x') = x'$. Then

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this contradicts contraction:

$$0 < d(f(x'), f(x_0)) = d(x', x_0) \not< d(x', x_0).$$

□

Example

$X = [0, \infty)$, $d_X(x, y) = |x - y|$. (X, d) is complete.

$$f : X \to X, \qquad x \longmapsto \lambda x + a, \quad a \geq 0,\ \lambda > 0.$$

f is contraction $\iff \lambda < 1$.

$$\lambda \geq 1 \begin{cases} a = 0: & 1 \text{ fixed point}, \\ a \neq 0: & 0 \text{ fixed points}, \end{cases} \qquad \lambda = 1,\ a = 0: \text{ all are fixed points.}$$

$\lambda < 1$: Unique fixed point.

Example (Counterexample)

Change $X = (0, \infty)$. Then f when $\lambda < 1$ has no fixed point.

Example

Let $I \subseteq \mathbb{R}$ be an interval. $f : I \to I$ differentiable with $\sup |f'(x)| < 1$.

By the Mean Value Theorem, f is contractive:

$$|f(x) - f(y)| \leq \sup |f'(x)|\, |x - y| < |x - y|\,.$$

Then, by the Banach Fixed-Point Theorem, f always has a unique fixed point in A.

5 Infinite Series

Definition

Let $(X, \|\cdot\|)$ be a normed vector space, $k_0 \in \mathbb{Z}$, and let $\{a_k\}_{k=k_0}^{\infty}$ be a sequence of elements in X. For every $n \geq k_0$, we call

$$S_n = \sum_{k=k_0}^{n} a_k$$

the partial sum of order n of $\{a_k\}_{k \geq k_0}$. If $\{S_n\}_{n=k_0}^{\infty}$ converges in $(X, \|\cdot\|)$ to $S \in X$, then we say

the formal series

$$\sum_{k=k_0}^{\infty} a_k$$

converges to S in $(X, \|\cdot\|)$ and its sum is equal to S. If $\{S_n\}_n$ diverges, we say $\sum_{k=k_0}^{\infty} a_k$ diverges.

Example

Geometric Series

$$\sum_{n=0}^{\infty} r^n, \qquad r \in \mathbb{R},$$

converges to $\dfrac{1}{1-r} \iff |r| < 1$. (Math 254)

Example (Neumann Series)

For every $A \in (M_{2\times 2}(\mathbb{R}), \|\cdot\|_{2,2})$, where

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$$\|A\|_{2,2} = \max_{\|x\|_2=1} \|Ax\|_2 ,$$

the condition $\|A\|_{2,2} < 1$ implies that the Neumann series converges and

$$\sum_{k=0}^{\infty} A^k = (\mathrm{Id} - A)^{-1}.$$

More generally, in finite dimensions the Neumann series converges if and only if the spectral radius $\rho(A)$ is less than 1. Thus $\|A\|_{2,2} < 1$ is sufficient, but it is not necessary. (Assignment 4)

Example

p-series:

$$\sum_{k=1}^{\infty} \frac{1}{k^p} \text{ converges} \iff p > 1.$$

(Assignment 5)

Proposition

$\forall k' \geq k_0$,

$$\sum_{k=k'}^{\infty} a_k \text{ converges} \iff \sum_{k=k_0}^{\infty} a_k \text{ converges}$$

Proof. Observe that one sequence of partial sums is simply shifted by a constant.

□

Proposition

If $\sum_{k=k_0}^{\infty} a_k$ converges, then $\lim_{k\to\infty} a_k = 0$.

Proof.

$$a_n = \sum_{k=k_0}^{n} a_k - \sum_{k=k_0}^{n-1} a_k \longrightarrow S - S = 0.$$

□

Proposition

If $\sum_{k=0}^{\infty} a_k$, $\sum_{k=0}^{\infty} b_k$ converge, then $\forall \lambda \in \mathbb{R}$,

$$\sum_{k=0}^{\infty} (\lambda a_k + b_k) = \lambda \sum_{k=0}^{\infty} a_k + \sum_{k=0}^{\infty} b_k.$$

Proof. This follows by looking at the sequence of partial sums. □

Proposition (Cauchy's convergence test)

If $(X, \|\cdot\|)$ is Banach, then

$$\sum_{k=k_0}^{\infty} a_k \text{ converges} \iff \text{Cauchy},$$

that is, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that, for every $n \geq m \geq N$,

$$\left\| \sum_{k=m}^{n} a_k \right\| < \varepsilon.$$

Proof. Trivial. Observe the partial sum is Cauchy, thus apply completeness.

□

Definition

We say $\sum_{k=k_0}^{\infty} a_k$ converges absolutely if

$$\sum_{k=k_0}^{\infty} \|a_k\| \text{ converges}$$

We say it converges conditionally if it converges but not absolutely.

Remark

By the monotone convergence theorem for sequences, we know

$$\sum_{k=k_0}^{\infty} \|a_k\| \text{ converges} \iff \text{the sequence of partial sums is bounded in } \mathbb{R}.$$

Then

$$\sum_{k=k_0}^{\infty} \|a_k\| = \sup_{n\in\mathbb{N}} \sum_{k=0}^{n} \|a_k\| .$$

Proposition

If $(X, \|\cdot\|)$ is Banach, then every absolutely convergent series is

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convergent.

Remark

In fact, $(X, \|\cdot\|)$ is a Banach space $\iff$ every absolutely convergent series converges in $(X, \|\cdot\|)$.

Proof. Exercise: first show that the sequence of partial sums is Cauchy using the triangle inequality. Then continue the argument. □

Proposition

Let $\sum_{k=k_0}^{\infty} a_k$ be an absolutely convergent series in a Banach space $(X, \|\cdot\|)$. Then every bijection

$$f : \{k \in \mathbb{Z} \mid k \geq k_0\} \to \{k \in \mathbb{Z} \mid k \geq k_0\},$$

$$\sum_{k=k_0}^{\infty} a_{f(k)}$$

converges and has same limit as $\sum_{k=k_0}^{\infty} a_k$.

(that is, not affected by rearrangement.)

Example

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k}.$$

By rearrangement, we can get a series

$$1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + \cdots$$

$$= \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \cdots$$
$$= \frac{1}{2}\left(\sum_{k=1}^{\infty} \frac{(-1)^k}{k}\right).$$

Proof. By absolute convergence, for every $\varepsilon > 0$ there exists $M \in \mathbb{N}$ such that

$$\sum_{k=M+1}^{\infty} \|a_k\| < \varepsilon.$$

Then

$$\begin{aligned}
\left\| \sum_{k=k_0}^{N} a_{f(k)} - \sum_{k=k_0}^{\infty} a_k \right\| &\leq \left\| \sum_{k=k_0}^{N} a_{f(k)} - \sum_{k=k_0}^{M} a_k \right\| + \left\| \sum_{k=M+1}^{\infty} a_k \right\| \\
&\leq \left\| \sum_{k=k_0}^{N} a_{f(k)} - \sum_{k=k_0}^{M} a_k \right\| + \sum_{k=M+1}^{\infty} \|a_k\| \\
&< \left\| \sum_{k=k_0}^{N} a_{f(k)} - \sum_{k=k_0}^{M} a_k \right\| + \varepsilon, \qquad \text{for all } M \leq N.
\end{aligned}$$

Define

$$N := \max_{k_0 \leq k < M} f^{-1}(k).$$

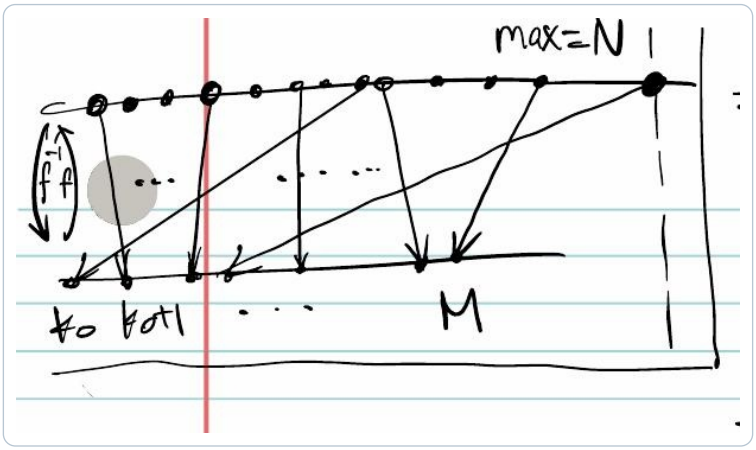
Then

$$f^{-1}(\{k_0, \ldots, M\}) \subseteq \{k_0, \ldots, N\},$$

and we have the extra terms in $f(\{k_0, \ldots, N\}) \setminus \{k_0, \ldots, M\}$ coming after $k = M$ in $\{a_k\}$. Thus

$$< \left\| \sum_{k=k_0}^{N} a_{f(k)} - \sum_{k=k_0}^{M} a_k \right\| + \varepsilon.$$

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$$= \left\| \text{extra terms not cancelled with } \sum_{k=k_0}^{M} a_k \right\| + \varepsilon \qquad \text{(must come after } k = M\text{)}$$

$$\leq \left\| \sum_{k=M+1}^{\infty} a_k \right\| + \varepsilon < \varepsilon + \varepsilon = 2\varepsilon.$$

□

5.1 Comparison Tests

Theorem

Let $\sum_{k=k_0}^{\infty} a_k$ and $\sum_{k=k_0}^{\infty} b_k$ be two series in $(\mathbb{R}, |\cdot|)$ such that

$$0 \leq a_k \leq b_k, \qquad \forall k.$$

Then

① If $\sum_{k=k_0}^{\infty} b_k$ converges, then $\sum_{k=k_0}^{\infty} a_k$ converges.

② If $\sum a_k$ diverges, then $\sum b_k$ diverges.

Proof. Use the monotone convergence theorem (for sequences) to prove ①; ② follows by contrapositive. Exercise. □

Theorem (Limit Comparison Test)

Let $\sum a_k$, $\sum b_k$ be two series in $(\mathbb{R}, |\cdot|)$ with strictly positive terms. Then if

$$\lim_{k\to\infty} \frac{a_k}{b_k} = l \in (0, +\infty),$$

then

$$\sum a_k \text{ converges} \iff \sum b_k \text{ converges}$$

Proof. Choose $\varepsilon = \frac{l}{2}$. There exists $N \in \mathbb{N}$ such that, for every $k \geq N$,

$$\frac{l}{2} < \left| \frac{a_k}{b_k} \right| < \frac{3l}{2}.$$

Then use the direct comparison test. □

Theorem (Root Test)

Let $\sum_{k=k_0}^{\infty} a_k$ be a series in $(\mathbb{R}, |\cdot|)$ such that $a_k \geq 0$, $\forall k \geq N$. Then

$$\limsup_{k\to\infty} \sqrt[k]{a_k} < 1 \implies \sum_{k=k_0}^{\infty} a_k \text{ converges.}$$

If

$$\limsup_{k\to\infty} \sqrt[k]{a_k} > 1, \quad \text{then } \sum a_k \text{ diverges.}$$

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If $= 1$, then it is inconclusive.

Proof. Let

$$r = \limsup \sqrt[k]{a_k}.$$

Suppose $r < 1$. Choose $\varepsilon > 0$ such that $r + \varepsilon < 1$. Then there exists $N \in \mathbb{N}$ such that $\forall k \geq N$,

$$|\sqrt[k]{a_k} - r| < \varepsilon \Rightarrow a_k < (r+\varepsilon)^k.$$

$\sum (r+\varepsilon)^k$ is a geometric series where $r + \varepsilon < 1 \Rightarrow$ it converges $\Rightarrow \sum a_k$ converges by the direct comparison test.

Now if $r > 1$, then by the properties of the limit superior,

$$\lim a_k \text{ (if existing)} = \limsup a_k > 1 \Rightarrow \text{divergence by the } n\text{th-term test.}$$

□

Example

$$a_k = \begin{cases} 2^{-k}, & \text{if } k \text{ is odd,} \\ 3^{-k}, & \text{if } k \text{ is even.} \end{cases}$$

Then

$$\limsup \sqrt[k]{a_k} = \max\left\{\frac{1}{2}, \frac{1}{3}\right\} = \frac{1}{2} < 1 \Rightarrow \text{convergence.}$$

Theorem (Ratio Test)

Let $\sum a_k$ be a series in $(\mathbb{R}, |\cdot|)$ with strictly positive terms. Then

① If $\limsup \dfrac{a_{k+1}}{a_k} < 1$, then $\sum a_k$ converges.

② If $\boxed{\liminf} \dfrac{a_{k+1}}{a_k} > 1$, then $\sum a_k$ diverges.

③ If limsup or liminf $= 1$, then inconclusive.

Remark

With the previous example, $\limsup \dfrac{a_{k+1}}{a_k} > 1$, yet $\sum a_k$ converges. Thus liminf in part ② cannot be replaced.

Proof. ① Assume

$$r = \limsup_{k\to\infty} \frac{a_{k+1}}{a_k} < 1.$$

Then, for every $\varepsilon \in (0, 1-r)$, there exists $N \in \mathbb{N}$ such that

$$\forall n \geq N, \qquad \sup_{l\geq n} \frac{a_{k+1}}{a_k} < r + \varepsilon.$$

Then bound $\sum a_k$ from above by a convergent geometric series.

② Bound a_k from below, then apply nth term test.

□

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Example

$$\sum_{k=0}^{\infty} \frac{|x|^k}{k!}$$

converges for all $x \in \mathbb{R}$ or $\mathbb{C}$.

(By Ratio test)

Thus we can define

$$\underline{\text{Exponential:}} \qquad e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

$$\underline{\text{Trigonometric functions:}} \qquad \sin x := \frac{e^{ix} - e^{-ix}}{2i}, \qquad \cos x = \frac{e^{ix} + e^{-ix}}{2},$$

$$\cosh x := \frac{e^x + e^{-x}}{2}, \qquad \sinh x = \frac{e^x - e^{-x}}{2}.$$

Theorem (Alternating Series Test)

Let

$$\sum_{k=k_0}^{\infty} (-1)^k a_k$$

be a series in $(\mathbb{R}, |\cdot|)$ such that

$$0 \leq a_{k+1} \leq a_k, \quad \forall k \geq k_0, \qquad \text{and} \qquad \lim_{k\to k_0} a_k = 0.$$

Then

$$\sum_{k=k_0}^{\infty} (-1)^k a_k \text{ converges} \qquad \text{and} \qquad 0 \leq \sum_{k=k_0}^{\infty} (-1)^{k-k_0} a_k \leq a_{k_0}.$$

Proof. Show S_{2n} is monotone decreasing and bounded below.

S_{2n+1} is monotone increasing and bounded above. $\Rightarrow$ both converge.

Show S_n is convergent by an ε–N argument. Exercise.

□

Proposition

Let $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ be series in $(\mathbb{R}, |\cdot|)$ such that $\sum a_k$ converges absolutely and $\sum b_k$ converges. Then for each n, define convolution between $\sum_{k=0}^{n} a_k$ and $\sum_{k=0}^{n} b_k$ as

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}.$$

Then $\sum_{n=0}^{\infty} c_n$ converges and

$$\sum_{n=0}^{\infty} c_n = \left(\sum_{k=0}^{\infty} a_k\right)\left(\sum_{k=0}^{\infty} b_k\right).$$

In particular, $\sum c_n$ is called the Cauchy product of $\sum a_k$ and $\sum b_n$.

Example

This proves $e^{x+y} = e^x e^y$ right away along using binomial expansion.

(Exercise)

Example (Counterexample)

Consider

$$a_k = b_k = \frac{(-1)^k}{\sqrt{k+1}}.$$

Then $\sum a_k$ and $\sum b_k$ converges. Yet

$$c_n = (-1)^n \sum_{k=0}^{n} \frac{1}{\sqrt{k+1}\sqrt{n-k+1}}$$

by calculations.

Notice that $(k+1)(n-k+1)$ attains max at $k = \frac{n}{2}$. Thus

$$|c_n| \geq \sum_{k=0}^{n} \frac{1}{\frac{n}{2}+1} = \frac{n}{\frac{n}{2}+1} = \frac{2n}{n+2}.$$

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Thus $c_n \nrightarrow 0 \Rightarrow \sum c_n$ can not converge.

Proof. $\forall N \in \mathbb{N}$,

$$\begin{aligned}\sum_{n=0}^{N} c_n &= \sum_{n=0}^{N}\left(\sum_{k=0}^{n} a_k b_{n-k}\right)\\ &= a_0b_0 + (a_0b_1 + a_1b_0) + \cdots + (a_Nb_0 + a_{N-1}b_1 + \cdots + a_0b_N)\\ &= a_0(b_0 + b_1 + \cdots + b_N) + a_1(b_0 + b_1 + \cdots + b_{N-1}) + \cdots + a_Nb_0\end{aligned}$$

$$
\begin{aligned}
&= \sum_{n=0}^{N} a_n \left(\sum_{k=0}^{N-n} b_k \right) \\
&= \sum_{n=0}^{N} a_n \left(\sum_{k=0}^{\infty} b_k - \sum_{k=N-n+1}^{\infty} b_k \right) \quad \text{since } \sum b_k \text{ converges} \\
&= \sum_{n=0}^{N} a_n \left(\sum_{k=0}^{\infty} b_k \right) - \sum_{n=0}^{N} a_n \left(\sum_{k=N-n+1}^{\infty} b_k \right) \\
&= \left(\sum_{k=0}^{\infty} b_k \right) \left(\sum_{n=0}^{N} a_n \right) - \sum_{n=0}^{N} a_n \left(\sum_{k=N-n+1}^{\infty} b_k \right).
\end{aligned}
$$

Thus to show our claim, the first term converges to $(\sum a_k)(\sum b_k)$; it is left to show the second term converges to 0 as $N \to \infty$.

$(*)$ Since $\sum_{k=0}^{\infty} b_k$ converges, for every $\varepsilon > 0$ there exists $M \in \mathbb{N}$ such that

$$\left| \sum_{k=M}^{\infty} b_k \right| < \varepsilon.$$

$(\dagger)$ Since $\sum a_k$ converges absolutely, $\lim_{k\to\infty} a_k = 0 \Rightarrow$ there exists $K \in \mathbb{N}$ such that

$$\forall k \geq K, \qquad |a_k| < \varepsilon.$$

Now compute $\forall N > K + M$,

$$
\begin{aligned}
\left| \sum_{n=0}^{N} a_n \left(\sum_{k=N-n+1}^{\infty} b_k \right) \right| &= \left| \sum_{n=0}^{N-M} a_n \left(\sum_{k=N-n+1}^{\infty} b_k \right) \right| + \left| \sum_{n=N-M+1}^{N} a_n \left(\sum_{k=N-n+1}^{\infty} b_k \right) \right| \\
&\leq \sum_{n=0}^{N-M} |a_n| \left| \sum_{k=N-n+1}^{\infty} b_k \right| + \left| \sum_{n=N-M+1}^{N} a_n \left(\sum_{k=N-n+1}^{\infty} b_k \right) \right| \\
&\leq \sum_{n=0}^{N-M} |a_k| \cdot \varepsilon \quad (\text{by } (*)) \\
&\quad + \underbrace{\left| \sum_{n=1}^{M} a_{(N-M)+n} \left(\sum_{k=M-n+1}^{\infty} b_k \right) \right|}_{\substack{\text{fixed } M \text{ terms} \\ (\#\text{ terms independent from } N)}} \quad \overset{\text{shift index}}{} \\
&\leq \varepsilon \sum_{n=0}^{\infty} |a_k| + \sum_{n=1}^{M} |a_{N-M+n}| \left| \sum_{k=M-n+1}^{\infty} b_k \right| \\
&\leq \varepsilon \sum_{n=0}^{\infty} |a_k| + \sum_{n=1}^{M} \varepsilon \left(\sum_{k=1}^{M} |b_k| + \left| \sum_{k=M+1}^{\infty} b_k \right| \right).
\end{aligned}
$$

Here $N - M \geq K$, $K + n > K$, and $1 \leq k \leq M$ in the indicated terms. The first bracket is bounded since $\sum a_k$ converges absolutely; the finite terms are bounded, and the last bracket is bounded since $\sum b_k$ converges. Say these bounds are B_1, B_2, so the last display is

$$\leq \varepsilon(B_1 + M \cdot B_2) = B \cdot \varepsilon.$$

Therefore this term converges to 0 as $N \to \infty$ (we can choose $\varepsilon' = \varepsilon/B$). $\square$

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6 Power Series

Definition

Let $\{a_k\}_{k=1}^{\infty}$ be a sequence of elements in $\mathbb{R}$ (or $\mathbb{C}$) and $x_0 \in \mathbb{R}$ or $\mathbb{C}$. The function

$$f(x) = \sum_{k=0}^{\infty} a_k (x - x_0)^k$$

defined on set where this series converges is called a power series.

Proposition

Let $\sum_{k=0}^{\infty} a_k (x - x_0)^k$ be a power series and

$$R = \frac{1}{\limsup_{k \to \infty} \sqrt[k]{|a_k|}}$$

with convention $\frac{1}{\infty} = 0$, $\frac{1}{0} = \infty$. Then $\forall x \in \mathbb{R}$ or $\mathbb{C}$,

$$|x - x_0| < R \Rightarrow \sum_{k=0}^{\infty} a_k (x - x_0)^k \text{ converges absolutely,}$$

$$|x - x_0| > R \Rightarrow \sum_{k=0}^{\infty} a_k (x - x_0)^k \text{ diverges.}$$

Lemma (Generalized Root test & Ratio test)

If $a_k \neq 0$, $\forall k$,

$$\liminf \frac{|a_{k+1}|}{|a_k|} > 1, \quad \text{then } \sum a_k \text{ diverges.}$$

If a_k is a sequence in $\mathbb{R}$,

$$\limsup_{k \to \infty} |a_k|^{\frac{1}{k}} > 1, \quad \text{then } \sum a_k \text{ diverges.}$$

(that is, a_k need not be positive.)

Proof. Recall in the proofs before, we have shown that $|a_n| > \varepsilon$, for some $\varepsilon > 0$, and thus nth term test failed.

Note that this furthermore implies that $a_n \nrightarrow 0$. Thus by nth term test, $\sum a_n$ diverges. □

Proof of the proposition. By root test, compute

$$\limsup_{k\to\infty} \left|a_k(x-x_0)^k\right|^{\frac{1}{k}} = |x-x_0| \limsup_{k\to\infty} |a_k|^{\frac{1}{k}} = \frac{1}{R}|x-x_0|.$$

If $|x-x_0| < R$, then $\frac{1}{R}|x-x_0| < 1 \Rightarrow$ formal power series converges.

If $|x-x_0| > R$, then $\frac{1}{R}|x-x_0| > 1 \Rightarrow$ formal power series diverges.

The edge cases ($R = \infty$ or 0) should be separately dealt with but they are trivial. □

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Example

$$\sum_{k=0}^{\infty} x^k : R = 1.$$

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} : R = \infty.$$

Definition

We call the R as above the radius of convergence of the power series.

Proposition

Let

$$f(x) = \sum_{k=0}^{\infty} a_k(x-x_0)^k$$

be a power series with radius of convergence $R > 0$. Then f is infinitely differentiable on $(x_0 - R, x_0 + R)$, and $\forall N \in \mathbb{N}$,

$$\begin{aligned} f^{(N)}(x) &= \left(\sum_{k=0}^{\infty} a_k(x-x_0)^k\right)^{(N)} \overset{*}{=} \sum_{k=0}^{\infty} \left(a_k(x-x_0)^k\right)^{(N)} \\ &= \sum_{k=0}^{\infty} k(k-1)\cdots(k-N+1)a_k(x-x_0)^{k-N}, \end{aligned}$$

$\forall x \in (x_0 - R, x_0 + R)$, with the same radius of convergence. In particular,

$$f^{(N)}(x_0) = N!\,a_N. \qquad \text{Therefore} \qquad a_N = \frac{f^{(N)}(x_0)}{N!}.$$

Proof. Simplification: Suppose $N = 1$. $N \geq 2$ follows by induction.

Suppose $x_0 = 0$. $x_0 \neq 0$ follows from shift: $x \longmapsto x - x_0$.

Then the series becomes

$$\sum_{k=1}^{\infty} k a_k (x-x_0)^{k-1}.$$

By the proposition, it has radius of convergence

$$R' = \limsup_{k\to\infty} \frac{1}{|ka_k|^{\frac{1}{k}}} \le \limsup_{k\to\infty} \frac{1}{|a_k|^{\frac{1}{k}}} = R.$$

We will prove later that $R' \ge R$.

Let $x \in (-R, R)$, $t \in \mathbb{R} \setminus \{0\}$, $n \ge 2$, then compute

$$(*) \quad \left| \frac{1}{t} \left(\sum_{k=0}^{n} a_k (x+t)^k - \sum_{k=0}^{n} a_k x^k \right) - \sum_{k=0}^{n} k a_k x^{k-1} \right|$$

Two limiting processes: $t \to 0$, $n \to \infty$. Are they independent?

$$\begin{aligned}
&= \left| \frac{1}{t} \sum_{k=0}^{n} a_k \left((x+t)^k - x^k - k x^{k-1} t \right) \right| \\
&= \left| \frac{1}{t} \sum_{k=0}^{n} a_k \left(k(x+s)^{k-1}(x+t-x) - k x^{k-1} t \right) \right| \qquad \text{by the Mean Value Theorem, } s \in (0,t) \\
&= \left| \sum_{k=0}^{n} k a_k \left((x+s)^{k-1} - x^{k-1} \right) \right| \\
&= \left| \sum_{k=2}^{n} k a_k (x+s')^{k-2} \cdot s' \cdot (k-1) \right| \qquad \text{by the Mean Value Theorem again, } s' \in (0,s) \\
&\le \sum_{k=2}^{n} k \left|a_k\right| \left(|x| + |s'|\right)^{k-2} \cdot |s'| \cdot (k-1) \\
&\le \sum_{k=2}^{n} k(k-1) \left|a_k\right| \left(|x| + |t|\right)^{k-2} \cdot |t| \qquad \text{since } 0 \le |s'| \le |s| \le |t| .
\end{aligned}$$

Since $|x| < R$, there exists $\widetilde{R}$ such that $|x| < \widetilde{R} < R$. Choose t such that

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$$t < \widetilde{R} - |x| .$$

Then

$$\le \sum_{k=2}^{n} k(k-1) \left|a_k\right| \widetilde{R}^{k-2} |t| .$$

Now observe that

$$\limsup \left| |a_k| \widetilde{R}^{k-2} \right|^{\frac{1}{k}} = \frac{\widetilde{R}}{R} < 1.$$

Thus there exist $\varepsilon > 0$ and $N \in \mathbb{N}$ such that, for every $k > N$,

$$|a_k| \widetilde{R}^{k-2} \le (1-\varepsilon)^k.$$

Then since

$$\begin{cases} k(k-1) \left|a_k\right| \widetilde{R}^{k-2} \le k(k-1)(1-\varepsilon)^k, \\ k \left|a_k\right| |x|^{k-1} \le k(1-\varepsilon)^k |x| , \end{cases}$$

and both series on the right converges by the ratio test, we have

① $\sum k a_k x^{k-1}$ converges absolutely, $x \in (-R, R) \Rightarrow R' = R$.

② $\sum k \left|a_k\right| |x|^{k-1}$ converges. $\Rightarrow$ we can send $n \to \infty$ in $(*)$, obtaining

$$\left|\frac{1}{t}\left(\sum_{k=0}^{\infty} a_k (x+t)^k - \sum_{k=0}^{\infty} a_k x^k\right) - \sum_{k=1}^{\infty} k a_k x^{k-1}\right| \leq \left(\sum_{k=2}^{\infty} k(k-1)\left|a_k\right| \widetilde{R}^{k-2}\right)|t|\,.$$

Then this is independent from n, we can send $t \to 0$, obtaining

$$\left|\frac{f(x+t)-f(x)}{t} - \sum_{k=1}^{\infty} k a_k x^{k-1}\right| \leq (\cdots)\,|t| \longrightarrow 0.$$

Therefore

$$f'(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}.$$

□

Note

Another way to prove this is to set up Riemann integration theory first, and along with the Weierstrass M-test justifying uniform convergence of the power series on $[x_0 - r, x_0 + r]$, $r < R$ the radius of convergence, we can prove more easily that the uniform limit of the derivatives is the derivative of the pointwise limit. See real analysis P.174.

Corollary

$f(x) = e^x \in C^\infty(\mathbb{R})$. $f'(x) = e^x$. f is increasing, bijective from $\mathbb{R}$ to $(0, \infty)$ and

$$\lim_{x \to \infty} e^x = \infty, \qquad \lim_{x \to -\infty} e^x = 0.$$

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Proof. Recall the definition

$$e^x := \sum \frac{x^n}{n!}.$$

Thus the first two points follow from the previous proposition.

Notice by the example on page 34, we have

$$e^x = \frac{1}{e^{-x}}.$$

Thus for $x \geq 0$, $e^x > 0$ by definition,

$$\text{for } x < 0, \quad e^x = \frac{1}{e^{-x}} > 0 \qquad \Longrightarrow \qquad f(\mathbb{R}) \subseteq (0, \infty).$$

f is increasing follows since $f'(x) = e^x > 0$, $\forall x \in \mathbb{R}$.

As a consequence, f is injective.

Moreover, $e^x \geq 1 + x$, $\forall x > 0$, thus

$$\lim_{x\to\infty} e^x \geq \lim_{x\to\infty} (1+x) = \infty.$$

$$\lim_{x\to-\infty} e^x = \lim_{x\to-\infty} \frac{1}{e^{-x}} = 0.$$

Lastly surjectivity follows from $\lim_{x\to\infty} f(x) = \infty$, $\lim_{x\to-\infty} f(x) = 0$, f is continuous, and the intermediate value theorem. □

7 Exponential and Logarithmic Functions

Corollary

By the previous corollary, we can define

$$\ln : (0,\infty) \longrightarrow \mathbb{R}$$

the inverse of e^x.

As well, $\forall p \in \mathbb{R}$,

$$p^x := e^{x \ln p}, \qquad \log_p x := \frac{\ln x}{\ln p}.$$

Proposition

$$\forall x \in (0,2), \qquad \ln x = \sum \frac{(-1)^{k-1}(x-1)^k}{k}.$$

Proof. By computing

$$\limsup_{k\to\infty} |a_k|^{1/k} = \limsup_{k\to\infty} \left(\frac{1}{k}\right)^{1/k} = \frac{1}{1} = 1,$$

we see the radius of convergence is

$$R = \frac{1}{\limsup |a_k|} = 1$$

for this power series.

Let

$$\widetilde{\ln} x := \sum_{k=1}^{\infty} \frac{(-1)^{k-1}(x-1)^k}{k}.$$

We want to show that $\widetilde{\ln} e^y = y$, $\forall y \in \mathbb{R}$.

By the previous proposition, we can differentiate $\widetilde{\ln} x$ term by term, with the derivative having the same radius of convergence $R = 1$. Then compute

$$\frac{d}{dy}[\widetilde{\ln} e^y - y] = e^y(\widetilde{\ln}' e^y) - 1,$$

$$\widetilde{\ln}' x = \sum_{k=1}^{\infty} (-1)^{k-1}(x-1)^{k-1}$$

$$= \sum_{k=0}^{\infty} (1-x)^k = \frac{1}{1-(1-x)} = \frac{1}{x}.$$

$$\implies \quad \frac{d}{dy}(\widetilde{\ln} e^y - y) = \frac{e^y}{e^y} - 1 = 0 \quad \implies \quad \widetilde{\ln} e^y - y = C, \quad \text{for } e^y \in (0,2),$$

C a constant.

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By picking $y = 0$, we see

$$\widetilde{\ln} e^y - y = \widetilde{\ln}(1) - 0 = 0 - 0 = 0$$

$$\implies C = 0, \qquad \widetilde{\ln} e^y = y \text{ when } e^y \in (0,2) \quad \implies \quad \widetilde{\ln} x = \ln x \text{ on } x \in (0,2).$$

□

7.1 Taylor Series

Definition

Let I be an interval in $\mathbb{R}$, $x_0 \in I$, $f : I \to \mathbb{R}$ be infinitely differentiable at x_0. Then the series

$$\sum_{k=0}^{\infty} \frac{1}{k!} f^{(k)}(x_0)(x - x_0)^k$$

is called the Taylor series of f at x_0.

Remark

By the last proposition, we know every power series

$$f(x) = \sum_{k=0}^{\infty} a_k (x - x_0)^k$$

is equal to its Taylor series on the interval of convergence since

$$a_k = \frac{f^{(k)}(x_0)}{k!}, \qquad \forall k \in \mathbb{N}.$$

However this is not true for a general infinitely differentiable function at x_0. For example,

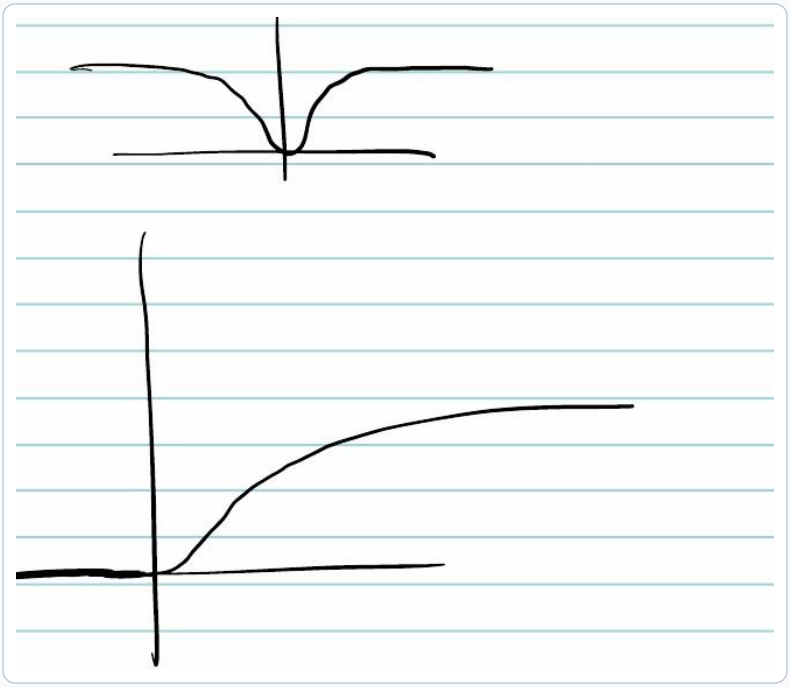
$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0 \end{cases}$$

is infinitely differentiable in $\mathbb{R}$

(Exercise: prove differentiability at $x = 0$ by induction, $f'(x) = 0$), yet its Taylor series diverges.

Similarly,

$$g(x) = \begin{cases} e^{-1/x}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$



Theorem (Taylor's theorem with Lagrange's form of remainder)

Let $x, x_0 \in \mathbb{R}$, $x \neq x_0$,

$$I = \begin{cases} [x, x_0], & \text{if } x < x_0, \\ [x_0, x], & \text{if } x > x_0, \end{cases} \qquad \mathring{I} = \begin{cases} (x, x_0), & \text{if } x < x_0, \\ (x_0, x), & \text{if } x > x_0, \end{cases}$$

and $f : I \to \mathbb{R} \in C^n(I)$, n^{th} continuously differentiable, and $(n+1)^{\text{th}}$ differentiable on

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$\mathring{I}$.

Then $\exists x' \in \mathring{I}$ such that

$$f(x) = \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(x - x_0)^k + \frac{1}{(n+1)!} f^{(n+1)}(x')(x - x_0)^{n+1}.$$

Proof. Define

$$M_{x,x_0} := \frac{f(x) - \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(x - x_0)^k}{(x - x_0)^{n+1}}.$$

We want to show that $\exists x' \in \mathring{I}$ such that

$$M_{x,x_0} = f^{(n+1)}(x') \cdot \frac{1}{(n+1)!}.$$

Define

$$g(t) := f(t) - \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(t - x_0)^k - M_{x,x_0}(t - x_0)^{n+1}.$$

Then g is n^{th}-order continuously differentiable, $(n+1)^{\text{th}}$-order differentiable on $\mathring{I}$. Compute

$$g(x_0) = g'(x_0) = \cdots = g^{(n)}(x_0) = 0.$$

Moreover, $g(x) = 0$ at $t = x$.

By Rolle's theorem / the mean value theorem,

$$g(x_0) = g(x) = 0 \Longrightarrow \exists \xi_1 \in \mathring{I} \text{ such that } g^{(1)}(\xi_1) = 0,$$
$$g^{(1)}(\xi_1) = g'(x_0) = 0 \Longrightarrow \exists \xi_2 \in \mathring{I} \text{ such that } g^{(2)}(\xi_2) = 0,$$
$$\vdots \qquad \text{Induction}$$
$$g^{(n)}(\xi_n) = g^{(n)}(x_0) = 0 \Longrightarrow \exists x' \in \mathring{I} \text{ such that } g^{(n+1)}(x') = 0.$$
$$\Longrightarrow \quad g^{(n+1)}(x') = f^{(n+1)}(x') - (n+1)! M_{x,x_0} = 0$$
$$\Longrightarrow \quad f^{(n+1)}(x') = (n+1)! M_{x,x_0}.$$

□

8 Riemann–Stieltjes Integration

Definition

Let $a < b \in \mathbb{R}$. A partition of $[a,b]$ is a finite subset $\{x_0, \ldots, x_n\}$, such that

$$a = x_0 \le x_1 \le \cdots \le x_{n-1} \le x_n = b.$$

Let $\alpha : [a,b] \to \mathbb{R}$ be non-decreasing, $f : [a,b] \to \mathbb{R}$ bounded. Then for every partition P of $[a,b]$, define

$$L(P,f,\alpha) := \sum_{i=1}^{n} m_i \Delta\alpha_i, \qquad U(P,f,\alpha) := \sum_{i=1}^{n} M_i \Delta\alpha_i,$$

where

$$m_i = \inf_{x \in [x_{i-1}, x_i]} f(x), \qquad M_i = \sup_{x \in [x_{i-1}, x_i]} f(x), \qquad \Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1}).$$

In the case where $\alpha = \mathrm{Id}$, we have the Riemann integrals.

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Define

$$\underline{\int_a^b} f \, d\alpha := \sup_{P \text{ partition}} L(P,f,\alpha), \qquad \overline{\int_a^b} f \, d\alpha := \inf_{P \text{ partition}} U(P,f,\alpha).$$

(Another notation is $\underline{\int_a^b} f(x) \, d\alpha(x)$.)

We call $\underline{\int_a^b} f \, d\alpha$ and $\overline{\int_a^b} f \, d\alpha$ the

lower and upper (Riemann–Stieltjes) integrals of f over $[a,b]$ with respect to α.

If

$$\underline{\int_a^b} f \, d\alpha = \overline{\int_a^b} f \, d\alpha,$$

we say f is Riemann–Stieltjes integrable. The Riemann–Stieltjes integral of f over $[a,b]$, with respect to α, is defined by

$$\int_a^b f\,d\alpha := \underline{\int_a^b} f\,d\alpha = \overline{\int_a^b} f\,d\alpha.$$

Lastly, denote

$$\mathcal{R}([a,b]) \text{ or } \mathcal{R}(\alpha) := \{\text{Riemann–Stieltjes integrable bounded } f \text{ on } [a,b] \text{ with respect to } \alpha\}.$$

Remark

1. Case where $\alpha = \mathrm{Id}$,
$$\int_a^b f\,d\alpha = \int_a^b f = \int_a^b f(x)\,d\alpha(x) = \int_a^b f(x)\,dx.$$
2. α is bounded since $\alpha(a) \le \alpha \le \alpha(b)$.
3. $$\underline{\int_a^b} f\,d\alpha,\ \overline{\int_a^b} f\,d\alpha \in [m(\alpha(b)-\alpha(a)), M(\alpha(b)-\alpha(a))].$$
4. * If α is differentiable on $[a,b]$, then
$$f \in \mathcal{R}([a,b],\alpha) \iff f\alpha' \text{ is Riemann integral.}$$
If this is the case, then
$$\int_a^b f\,d\alpha = \int_a^b f\alpha'\,dx.$$

Definition

Given partitions P^*, P of $[a,b]$, P^* is a refinement of P if $P \subseteq P^*$.

We call $P^{**} := P^* \cup P$ the common refinement of P and P^*.

Proposition

If P^* is a refinement of P, then

$$L(P^*,f,\alpha) \ge L(P,f,\alpha), \qquad U(P^*,f,\alpha) \le U(P,f,\alpha).$$

Proof. We prove it for the lower sum, upper sum follows.

We also assume P^* has one extra point than P. General case by induction.

Suppose it is in $[x_{i-1}, x_i]$ such that $\exists x^* \in (x_{i-1}, x_i)$ in P^*.

Then

$$\begin{aligned} L(P^*, f, \alpha) - L(P, f, \alpha) &= \inf_{[x_{i-1}, x^*]} f \cdot (\alpha(x^*) - \alpha(x_{i-1})) \\ &\quad + \inf_{[x^*, x_i]} f \cdot (\alpha(x_i) - \alpha(x^*)) \\ &\quad - \inf_{[x_{i-1}, x_i]} f \, (\alpha(x_i) - \alpha(x_{i-1})) \\ &\geq \inf_{[x_{i-1}, x_i]} f \, (\alpha(x^*) - \alpha(x_{i-1}) \\ &\qquad\qquad + \alpha(x_i) - \alpha(x^*) \\ &\qquad\qquad - \alpha(x_i) + \alpha(x_{i-1})) = 0. \end{aligned}$$

□

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Proposition

$$\underline{\int_a^b} f \, d\alpha \leq \overline{\int_a^b} f \, d\alpha.$$

Proof. For every pair of partitions P_1, P_2 of $[a, b]$, by the previous proposition,

$$L(P_1, f, \alpha) \leq L(P_1 \cup P_2, f, \alpha) \leq U(P_1 \cup P_2, f, \alpha) \leq U(P_2, f, \alpha).$$

Then taking sup among P_1 on left, inf on P_2 on right, we have

$$\underline{\int_a^b} f \, d\alpha \leq \overline{\int_a^b} f \, d\alpha.$$

□

Proposition (Riemann Integrability Criterion)

$$f \in \mathcal{R}([a, b], \alpha) \iff \forall \varepsilon > 0, \ \exists P \text{ partition of } [a, b] \text{ such that}$$
$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon.$$

Proof. This is good practice. The reader could attempt it before reading the proof.

($\Rightarrow$) Suppose $f \in \mathcal{R}([a, b], \alpha)$. Let $\varepsilon > 0$.

Then

$$\underline{\int_a^b} f \, d\alpha = \int_a^b f \, d\alpha = \overline{\int_a^b} f \, d\alpha,$$
$$\sup_P L(P, f, \alpha), \qquad \inf_P U(P, f, \alpha).$$

$$\implies \exists P_1 \text{ partition such that } \int_a^b f \, d\alpha - \frac{\varepsilon}{2} < L(P_1, f, \alpha) \leq \int_a^b f \, d\alpha,$$

$$\exists P_2 \text{ partition such that } \int_a^b f \, d\alpha \leq U(P_2, f, \alpha) < \int_a^b f \, d\alpha + \frac{\varepsilon}{2}.$$

For common refinement $P = P_1 \cup P_2$,

$$\int_a^b f\,d\alpha - \frac{\varepsilon}{2} < L(P,f,\alpha) \le \int_a^b f\,d\alpha,$$

$$\int_a^b f\,d\alpha \le U(P,f,\alpha) < \int_a^b f\,d\alpha + \frac{\varepsilon}{2}.$$

$$\Longrightarrow U(P,f,\alpha) - L(P,f,\alpha) < \int_a^b f\,d\alpha + \frac{\varepsilon}{2} - \int_a^b f\,d\alpha + \frac{\varepsilon}{2} = \varepsilon.$$

($\Leftarrow$) Notice that

$$\underline{\int_a^b} f\,d\alpha \ge L(P,f,\alpha),\ \forall P, \qquad \overline{\int_a^b} f\,d\alpha \le U(P,f,\alpha),\ \forall P \quad \text{by assumption.}$$

Thus

$$\overline{\int_a^b} f\,d\alpha - \underline{\int_a^b} f\,d\alpha \le U(P,f,\alpha) - L(P,f,\alpha) < \varepsilon, \quad \forall \varepsilon > 0.$$

$$\Longrightarrow \quad \overline{\int_a^b} f\,d\alpha = \underline{\int_a^b} f\,d\alpha.$$

□

Example (Assignment 6, Question 2)

If α is constant on $[a,s)$ and $(s,b]$, for $s \in [a,b]$, and f is bounded on $[a,b]$ and continuous at s, then

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$$f \in \mathcal{R}([a,b],\alpha) \qquad \text{and} \qquad \int_a^b f\,d\alpha = f(s)(\alpha(b) - \alpha(a)).$$

Example (Counterexample: Dirichlet's function)

$$f(x) = \begin{cases} 0, & x \in [a,b] \cap \mathbb{Q}, \\ 1, & x \in [a,b] \cap \mathbb{Q}^c. \end{cases}$$

Since in every partition, $m_i = 0$, $M_i = 1$ ($\mathbb{Q}$ is dense in $\mathbb{R}$), we have

$$L(P,f,\alpha) = 0, \qquad U(P,f,\alpha) = \sum \Delta\alpha_i = \alpha(b) - \alpha(a).$$

Hence when α is nonconstant, f is not Riemann–Stieltjes integrable.

Proposition

Let $f, \alpha : [a,b] \longrightarrow \mathbb{R}$ such that f is bounded on $[a,b]$, α is increasing on $[a,b]$. Let D be the set of points in $[a,b]$ at which f is not continuous. Then if D is countable and α is continuous on D.

Then
$$f \in \mathcal{R}([a,b],\alpha).$$

Proof. Let $\varepsilon > 0$. $\forall x \in [a,b] \setminus D$, since f is continuous at x, $\exists \delta_{\varepsilon,x} > 0$ such that
$$(*) \qquad \forall y \in (x-\delta_{\varepsilon,x}, x+\delta_{\varepsilon,x}) \cap [a,b], \qquad |f(x)-f(y)| < \varepsilon.$$
Since D is countable, we can write $D = \{x_k\}_{k=1}^{\infty}$.
Since α is continuous on D, $\exists \delta_{\varepsilon,x_k} > 0$ such that
$$\forall y \in (x-\delta_{\varepsilon,x_k}, x+\delta_{\varepsilon,x_k}) \cap [a,b], \qquad |\alpha(x_k)-\alpha(y)| < \frac{1}{2^k}\varepsilon.$$
Since
$$[a,b] \subseteq \bigcup_{x\in[a,b]} (x-\delta_{\varepsilon,x}, x+\delta_{\varepsilon,x}),$$
$\exists y_1,\dots,y_N \in [a,b]$ such that
$$[a,b] \subseteq \bigcup_{i=1}^{N} (y_i-\delta_{\varepsilon,y_i}, y_i+\delta_{\varepsilon,y_i}) \qquad \text{by topological compactness.}$$
By ordering the endpoints in this finite union, we obtain a partition of $[a,b]$. Call it $P = \{x_0,\dots,x_n\}$. Then clearly $\forall j \in \{x_0,\dots,x_n\}$, $\exists i_j \in \{1,\dots,N\}$ such that
$$(x_{j-1}, x_j) \subseteq (y_{i_j}-\delta_{\varepsilon,y_{i_j}}, y_{i_j}+\delta_{\varepsilon,y_{i_j}}) \cap [a,b].$$

Case $y_{i_j} \in D$:
$$M_j - m_j \le 2B \qquad \text{since } f \text{ is bounded,}$$
but
$$\Delta\alpha_j = \alpha(x_j)-\alpha(x_{j-1}) \le \alpha(y_{i_j}+\delta_{\varepsilon,y_{i_j}}) - \alpha(y_{i_j}-\delta_{\varepsilon,y_{i_j}}) < \frac{1}{2^{k-1}}\varepsilon.$$

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Case $y_{i_j} \notin D$:
$$M_j - m_j \le f(y_{i_j}) + \varepsilon - (f(y_{i_j})-\varepsilon) = 2\varepsilon, \qquad \text{by } (*).$$
Thus
$$\begin{aligned} U(P,f,\alpha) - L(P,f,\alpha) &= \sum_{j=1}^{n} (M_j-m_j)\Delta\alpha_j \\ &\le \underbrace{\sum_{k=1}^{\infty} 2B\cdot\frac{1}{2^{k-1}}\varepsilon}_{y_{i_j}\in D} + \underbrace{2\varepsilon\cdot\sum_{j=1}^{n}\Delta\alpha_j}_{y_{i_j}\notin D} \\ &= 4B\cdot\varepsilon + 2\varepsilon\cdot(\alpha(b)-\alpha(a)) \\ &= (4B+2(\alpha(b)-\alpha(a)))\varepsilon. \end{aligned}$$
By the Riemann integrability criterion, f is Riemann–Stieltjes integrable. □

We recall two propositions from Mathematics 254.

$$\boxed{\text{Define } f(x\pm) := \lim_{t\to x^{\pm}} f(t).}$$

Proposition (I)

Suppose $f : (a, b) \to \mathbb{R}$ is monotone increasing. Then $f(x+)$, $f(x-)$ exists $\forall x \in (a, b)$.

Furthermore,

$$\sup_{a<t<x} f(t) = f(x-) \leq f(x) \leq f(x+) = \inf_{x<t<b} f(t),$$

and $f(x+) \leq f(y-)$, $\forall\, a < x < y < b$.

Proof.

$$\sup_{a<t<x} f(t) = f(x-), \qquad f(x+) = \inf_{x<t<b} f(t)$$

follows from the definition.

Their existence follows since

$$\{f(t) \mid x < t < b\}, \qquad \text{respectively,} \qquad \{f(t) \mid a < t < x\}$$

is bounded below (respectively, above) by $f(x)$ by monotonicity of f.

Lastly, since $x < y$,

$$\begin{aligned} f\left(\frac{x+y}{2}\right) &\in \{f(t) \mid a < t < y\} \\ &\in \{f(t) \mid x < t < b\} \\ \implies \quad f(x+) = \inf_{x<t<b} f(t) \leq f\left(\frac{x+y}{2}\right) &\leq \sup_{a<t<y} f(t) = f(y-). \end{aligned}$$

□

Proposition (II)

If $f : [a, b] \to \mathbb{R}$ is monotone increasing, then it has at most countably many points of discontinuity.

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Proof. Let $d \in (a, b)$ be a discontinuity of f. Then we must have $f(d-) < f(d+)$ (otherwise contradicting the monotonicity of f).

By density of $\mathbb{Q}$ in $\mathbb{R}$, we can map d to some

$$q \in \mathbb{Q} \cap (f(d-), f(d+)).$$

Furthermore, this map is injective by

$$f(x+) \leq f(y-), \qquad \forall\, a < x < y < b.$$

Thus

$$|D| \leq |\mathbb{Q}| = |\mathbb{N}|.$$

□

Proposition

Let $[a,b],[c,d] \subseteq \mathbb{R}$, $\alpha : [a,b] \to \mathbb{R}$ increasing on $[a,b]$, $f : [a,b] \to [c,d]$, $\varphi : [c,d] \to \mathbb{R}$ continuous on $[c,d]$. Then if

$$f \in \mathcal{R}([a,b],\alpha),$$

then

$$\varphi \circ f \in \mathcal{R}([a,b],\alpha).$$

Proof. By the Heine–Cantor theorem, φ is uniformly continuous, thus $\exists \delta > 0$ such that

$$\forall x, y \in [c,d], \quad |x-y| < \delta, \quad \text{we have} \quad |\varphi(x) - \varphi(y)| < \lambda\varepsilon,$$

for some $\lambda \in (0,\infty)$ to be decided.

Since $f \in \mathcal{R}([a,b],\alpha)$, $\exists P = \{x_0, \dots, x_n\}$ partition of $[a,b]$ such that

$$U(P,f,\alpha) - L(P,f,\alpha) < \frac{\theta\varepsilon\delta}{2}, \qquad \text{that is,} \qquad \sum_{i=1}^{n}(M_i - m_i)\Delta\alpha_i < \frac{\theta\varepsilon\delta}{2},$$

for some $\theta \in (0,\infty)$ to be determined. Then notice

$$U(P,\varphi\circ f,\alpha) - L(P,\varphi\circ f,\alpha) = \sum_{i=1}^{n}\left(\begin{array}{l}\sup\limits_{x_{i-1}\le x\le x_i} \varphi\circ f(x) \\ -\inf\limits_{x_{i-1}\le x\le x_i} \varphi\circ f(x)\end{array}\right)\Delta\alpha_i.$$

Define

$$\omega_i := M_i - m_i, \qquad \omega_i' := \sup_{x_{i-1}\le x\le x_i} \varphi\circ f(x) - \inf_{x_{i-1}\le x\le x_i} \varphi\circ f(x).$$

Then

$$\omega_i' \le \begin{cases} \lambda\varepsilon, & \text{if } \omega_i < \delta, \\ c := \sup\limits_{c\le x\le d}\varphi(x) - \inf\limits_{c\le x\le d}\varphi(x), & \text{if } \omega_i \ge \delta. \end{cases}$$

It follows that

$$\begin{aligned} U(P,\varphi\circ f,\alpha) - L(P,\varphi\circ f,\alpha) &\le \lambda\varepsilon \sum_{\substack{i\in\{1,\dots,n\}\\ \omega_i<\delta}} \Delta\alpha_i + c \sum_{\substack{i\in\{1,\dots,n\}\\ \omega_i\ge\delta}} \Delta\alpha_i \\ &\le \lambda\varepsilon\sum_{i=1}^{n}\Delta\alpha_i + c\sum_{\substack{i\in\{1,\dots,n\}\\ \omega_i\ge\delta}} \frac{\omega_i}{\delta}\Delta\alpha_i. \end{aligned}$$

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$$\begin{aligned} &\le \lambda\varepsilon(\alpha(b)-\alpha(a)) + c\cdot\sum_{i=1}^{n}\frac{\omega_i}{\delta}\Delta\alpha_i \\ &= \lambda\varepsilon(\alpha(b)-\alpha(a)) + \frac{c}{\delta}\sum_{i=1}^{n}\omega_i\Delta\alpha_i \\ &\le \lambda\varepsilon(\alpha(b)-\alpha(a)) + \frac{c}{\delta}\frac{\theta\varepsilon\delta}{2} = \lambda\varepsilon(\alpha(b)-\alpha(a)) + c\theta\frac{\varepsilon}{2}. \end{aligned}$$

Choose

$$\lambda := \frac{1}{2(\alpha(b) - \alpha(a))}, \qquad \theta := \frac{1}{c}$$

(these definitions are independent of the choice of δ and partition P), thus

$$U(P, \varphi \circ f, \alpha) - L(P, \varphi \circ f, \alpha) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

Proposition

Let $[a, b] \subseteq \mathbb{R}$, $\alpha : [a, b] \to \mathbb{R}$ increasing on $[a, b]$. Then

Linearity

(i) $\forall f, g \in \mathcal{R}([a, b], \alpha)$,

$$f + g \in \mathcal{R}([a, b], \alpha) \quad \text{and} \quad \int_a^b (f + g)\, d\alpha = \int_a^b f\, d\alpha + \int_a^b g\, d\alpha.$$

(ii) $\forall f \in \mathcal{R}([a, b], \alpha)$, $\lambda \in \mathbb{R}$,

$$\lambda f \in \mathcal{R}([a, b], \alpha) \quad \text{and} \quad \int_a^b \lambda f\, d\alpha = \lambda \int_a^b f\, d\alpha.$$

monotonicity

(iii) $\forall f, g \in \mathcal{R}([a, b], \alpha)$, $f \leq g$

$$\implies \quad \int_a^b f\, d\alpha \leq \int_a^b g\, d\alpha.$$

Δ-inequality

(iv) $\forall f \in \mathcal{R}([a, b], \alpha)$,

$$|f| \in \mathcal{R}([a, b], \alpha) \quad \text{and} \quad \left| \int_a^b f\, d\alpha \right| \leq \int_a^b |f|\, d\alpha.$$

(v) $\forall f, g \in \mathcal{R}([a, b], \alpha)$,

$$fg \in \mathcal{R}([a, b], \alpha).$$

Additivity

(vi) $\forall f \in \mathcal{R}([a, b], \alpha)$, $s \in (a, b)$,

$$f \in \mathcal{R}([a, s], \alpha), \qquad f \in \mathcal{R}([s, b], \alpha),$$

and

$$\int_a^b f\, d\alpha = \int_a^s f\, d\alpha + \int_s^b f\, d\alpha.$$

Proof. (i) Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. First notice that

$$\sup_{x_{i-1} \le x \le x_i} (f+g)(x) \le \sup_{x_{i-1} \le x \le x_i} f(x) + \sup_{x_{i-1} \le x \le x_i} g(x),$$

and

$$\inf_{x_{i-1} \le x \le x_i} (f+g)(x) \ge \inf_{x_{i-1} \le x \le x_i} f(x) + \inf_{x_{i-1} \le x \le x_i} g(x).$$

(exercise for Mathematics 254). Thus it follows by definition that

$$U(P, f+g, \alpha) \le U(P, f, \alpha) + U(P, g, \alpha),$$

$$L(P, f+g, \alpha) \ge L(P, f, \alpha) + L(P, g, \alpha).$$

$$\begin{aligned} \implies \quad & \sup_P L(P, f, \alpha) + \sup_P L(P, g, \alpha) \le \sup_P L(P, f+g, \alpha) \\ & \le \inf_P U(P, f+g, \alpha) \le \inf_P U(P, f, \alpha) + \inf_P U(P, g, \alpha) \end{aligned}$$

$$\begin{aligned} \implies \quad \int_a^b f\,d\alpha + \int_a^b g\,d\alpha &\le \underline{\int_a^b} (f+g)\,d\alpha \\ &\le \overline{\int_a^b} (f+g)\,d\alpha \\ &\le \int_a^b f\,d\alpha + \int_a^b f\,d\alpha. \end{aligned}$$

$$\implies \quad \text{“=”} \qquad \text{“=”}.$$

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(ii) If $\lambda \ge 0$, then

$$\sup \lambda f = \lambda \sup f, \qquad \inf \lambda f = \lambda \inf f$$

$$\implies \quad U(P, \lambda f, \alpha) = \lambda U(P, f, \alpha), \qquad L(P, \lambda f, \alpha) = \lambda L(P, f, \alpha), \quad \forall P.$$

$$\implies \quad \overline{\int_a^b} \lambda f\,d\alpha = \lambda \overline{\int_a^b} f\,d\alpha, \qquad \underline{\int_a^b} \lambda f\,d\alpha = \lambda \underline{\int_a^b} f\,d\alpha.$$

If $\lambda < 0$: Notice

$$\sup(-f) = -\inf f \quad \implies \quad U(P, -f, \alpha) = -L(P, f, \alpha),$$

$$L(P, -f, \alpha) = -U(P, f, \alpha)$$

$$\implies \quad \int_c^b (-f)\,d\alpha = -\int_c^b f\,d\alpha.$$

Then apply what we showed to

$$\int_a^b -\underbrace{(-\lambda)}_{>0} f\,d\alpha.$$

(iii) By (i), (ii), it suffices to show that $\int_a^b f\,d\alpha \ge 0$ for $f \ge 0$. But this is trivially true since

$$f \ge 0 \implies L(P, f, \alpha) \ge 0, \quad \forall P \implies \underline{\int_a^b} f\,d\alpha \ge 0.$$

Hence for $f \in \mathcal{R}([a,b],\alpha)$,

$$\int_a^b f\,d\alpha = \underline{\int_a^b} f\,d\alpha \geq 0.$$

Then apply (i), (ii) and this point to

$$\int_a^b (f-g)\,d\alpha.$$

(iv) Notice that $f \leq |f|$, $-f \leq |f|$, thus

$$\int_a^b f\,d\alpha \leq \int_a^b |f|\,d\alpha \quad \text{and} \quad \int_a^b -f\,d\alpha = -\int_a^b f\,d\alpha \leq \int_a^b |f|\,d\alpha$$

$$\implies \quad \left|\int_a^b f\,d\alpha\right| \leq \int_a^b |f|\,d\alpha.$$

(v) Notice that

$$fg = \frac{1}{4}\{(f+g)^2 - (f-g)^2\},$$

and $\varphi(x) = x^2$ is continuous, hence by the previous proposition,

$$\underbrace{(f+g)^2}_{\varphi\circ(f+g)}, \qquad \underbrace{(f-g)^2}_{\varphi\circ(f-g)} \in \mathcal{R}([a,b],\alpha) \implies fg \in \mathcal{R}([a,b],\alpha)$$

by (i) and (ii).

(vi) Let $\varepsilon > 0$. Since $f \in \mathcal{R}([a,b],\alpha)$, $\exists P = \{x_0,\ldots,x_n\}$ partition of $[a,b]$, such that

$$U(P,f,\alpha) - L(P,f,\alpha) < \varepsilon.$$

If $s \notin P$, define P' to be the refined partition

$$\{x_0,\ldots,x_i,s,x_{i+1},x_n\},$$

and suppose

$$x_0 \leq x_1 \leq \cdots \leq x_i \leq s \leq x_{i+1} \leq \cdots \leq x_n.$$

Then

$$P_1 = \{x_0,\ldots,x_i,s\}$$

is a partition of $[a,s]$, and

$$P_2 = \{s,x_{i+1},\ldots,x_n\}$$

is a partition of $[s,b]$.

Moreover, by definition, we know

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$$U(P',f,\alpha) = U(P_1,f,\alpha) + U(P_2,f,\alpha),$$
$$L(P',f,\alpha) = L(P_1,f,\alpha) + L(P_2,f,\alpha).$$

By a previous proposition,

$$U(P',f,\alpha) \leq U(P,f,\alpha), \qquad L(P',f,\alpha) \geq L(P,f,\alpha)$$

since P' is a refinement of P.

Compute

$$\begin{aligned} U(P', f, \alpha) - L(P', f, \alpha) &= U(P_1, f, \alpha) - L(P_1, f, \alpha) \\ &\quad + U(P_2, f, \alpha) - L(P_2, f, \alpha) \\ &\leq U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon. \end{aligned}$$

Furthermore, $U(P_i, f, \alpha) \geq L(P_i, f, \alpha)$ by another previous proposition. Hence

$$\underbrace{U(P_1, f, \alpha) - L(P_1, f, \alpha)}_{\geq 0} + \underbrace{U(P_2, f, \alpha) - L(P_2, f, \alpha)}_{\geq 0} < \varepsilon$$

$$\implies \quad U(P_i, f, \alpha) - L(P_i, f, \alpha) < \varepsilon$$

$$\implies \quad f \in \mathcal{R}([a, s], \alpha) \text{ and } f \in \mathcal{R}([s, b], \alpha)$$

by Riemann integrability criterion.

Then choose P such that

$$U(P, f, \alpha) < \int_a^b f \, d\alpha + \varepsilon \qquad \text{and} \qquad L(P, f, \alpha) > \int_a^b f \, d\alpha - \varepsilon,$$

and defining $P' := P \cup \{s\}$, P_1, P_2 as before, we have

$$\begin{aligned} \int_a^b f \, d\alpha - \varepsilon < L(P, f, \alpha) &\leq L(P', f, \alpha) \\ &\leq \int_a^s f \, d\alpha + \int_s^b f \, d\alpha \\ &\leq U(P', f, \alpha) \leq U(P, f, \alpha) < \int_a^b f \, d\alpha + \varepsilon. \end{aligned}$$

Sending $\varepsilon \to 0$, we have

$$\int_a^b f \, d\alpha = \int_a^s f \, d\alpha + \int_s^b f \, d\alpha.$$

□

Proposition

Let $[a, b] \subseteq \mathbb{R}$, $\alpha, \alpha_1, \alpha_2 : [a, b] \to \mathbb{R}$ be increasing on $[a, b]$. Then

(i) $\forall f \in \mathcal{R}([a, b], \alpha_1) \cap \mathcal{R}([a, b], \alpha_2)$,

$$f \in \mathcal{R}([a, b], \alpha_1 + \alpha_2),$$

and

$$\int_a^b f \, d(\alpha_1 + \alpha_2) = \int_a^b f \, d\alpha_1 + \int_a^b f \, d\alpha_2.$$

(ii) $\forall f \in \mathcal{R}([a, b], \alpha)$, $\lambda \in \mathbb{R}$,

$$f \in \mathcal{R}([a, b], \lambda\alpha)$$

and

$$\int_a^b f \, d(\lambda\alpha) = \lambda \int_a^b f \, d\alpha.$$

Proof.

$$\Delta(\alpha_1+\alpha_2)_i = (\alpha_1+\alpha_2)(x_i) - (\alpha_1+\alpha_2)(x_{i-1})$$

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$$\begin{aligned}
&= \alpha_1(x_i) - \alpha_1(x_{i-1}) + \alpha_2(x_i) - \alpha_2(x_{i-1}) \\
&= \Delta(\alpha_1)_i + \Delta(\alpha_2)_i. \\
\Delta(\lambda\alpha)_i &= (\lambda\alpha)(x_i) - (\lambda\alpha)(x_{i-1}) \\
&= \lambda\alpha(x_i) - \lambda\alpha(x_{i-1}) \\
&= \lambda\Delta\alpha_i.
\end{aligned}$$

Then everything follows from the definition. Exercise. □

8.1 Fundamental Theorem of Calculus

Theorem (Fundamental Theorem of Calculus part 1)

Let $[a,b] \subseteq \mathbb{R}$, $f \in \mathcal{R}([a,b])$ (that is, Riemann integrable). Define $F : [a,b] \to \mathbb{R}$ via

$$F(x) := \int_a^x f.$$

Then F is Lipschitz continuous on $[a,b]$. If f is continuous at $x_0 \in [a,b]$, then F is differentiable at x_0 with $F'(x_0) = f(x_0)$.

Proof. $\forall x, y \in [a,b]$, $x < y$,

$$|F(x) - F(y)| = \left| \int_a^x f - \int_a^y f \right| = \left| \int_y^x f \right| \leq \sup_{x \leq t \leq y} |f(t)|(y-x).$$

[f is bounded simply by the definition of Riemann–Stieltjes integrals.]

If f is continuous at $x_0 \in [a,b]$,

$$\begin{aligned}
\left| \frac{F(x_0) - F(x)}{x_0 - x} - f(x_0) \right| &= \left| \frac{\int_x^{x_0} f}{x_0 - x} - f(x_0) \right| \\
= \left| \frac{\int_x^{x_0} f(t)\,dt}{x_0 - x} - \frac{\int_x^{x_0} f(x_0)\,dt}{x_0 - x} \right| &= \left| \frac{\int_x^{x_0} [f(t) - f(x_0)]\,dt}{x_0 - x} \right| \\
\leq \sup_{t \in [x_0,x] \text{ or } [x,x_0]} |f(t) - f(x_0)| \cdot |x_0 - x| \cdot \frac{1}{|x_0 - x|} &= \sup |f(t) - f(x_0)|.
\end{aligned}$$

Let $\varepsilon > 0$. Choose $\delta > 0$ such that

$$|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon.$$

Thus

$$\left| \frac{F(x_0) - F(x)}{x_0 - x} - f(x_0) \right| \leq \sup_{[x,x_0] \text{ or } [x_0,x]} |f(t) - f(x_0)| \leq \varepsilon,$$

$\forall x$ such that $|x - x_0| < \delta$, hence

$$\lim_{x \to x_0} \frac{F(x_0) - F(x)}{x_0 - x} = f(x_0) \qquad \| \; F'(x_0).$$

□

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Theorem (Fundamental Theorem of Calculus part 2)

Let $[a,b] \subseteq \mathbb{R}$. $F : [a,b] \to \mathbb{R}$ such that F is continuous on $[a,b]$, differentiable on (a,b), and

$$F' = f \in \mathcal{R}([a,b]).$$

Then

$$\int_a^b f = F(b) - F(a).$$

Proof. Let $\varepsilon > 0$. By the Riemann integrability criterion, $\exists$ partition P such that

$$U(P,f,\alpha) - L(P,f,\alpha) < \varepsilon, \qquad P = \{x_0, x_1, \dots, x_n\}.$$

By the mean value theorem, $\exists \xi_i \in (x_{i-1}, x_i)$ such that

$$F'(\xi_i)\Delta x_i = F(x_i) - F(x_{i-1}), \qquad \| \qquad f(\xi_i)\Delta x_i.$$

Then compute

$$\begin{aligned}
\left| \int_a^b f - (F(b) - F(a)) \right| &= \left| \sum_{i=1}^n \int_{x_{i-1}}^{x_i} f - \sum_{i=1}^n (F(x_i) - F(x_{i-1})) \right| \\
&= \left| \sum_{i=1}^n \int_{x_{i-1}}^{x_i} f - \sum_{i=1}^n f(\xi_i)\Delta x_i \right| \\
&= \left| \sum_{i=1}^n \int_{x_{i-1}}^{x_i} f - \sum_{i=1}^n \int_{x_{i-1}}^{x_i} f(\xi_i) \right| \\
&\le \sum_{i=1}^n \int_{x_{i-1}}^{x_i} |f(t) - f(\xi_i)| \, dt \\
&\le \sum_{i=1}^n \int_{x_{i-1}}^{x_i} (M_i - m_i) \\
&= \sum_{i=1}^n (M_i - m_i)\Delta x_i \\
&= \sum_{i=1}^n M_i \Delta \alpha_i - \sum_{i=1}^n m_i \Delta \alpha_i \\
&= U(P,f,\alpha) - L(P,f,\alpha) < \varepsilon.
\end{aligned}$$

Sending $\varepsilon \to 0$ gives

$$\int_a^b f = F(b) - F(a).$$

□

Corollary (Integration by Parts)

Let $[a,b] \subseteq \mathbb{R}$, $F, G : [a,b] \to \mathbb{R}$ continuous on $[a,b]$, differentiable on (a,b), and

$$F' = f \in \mathcal{R}([a,b]), \qquad G' = g \in \mathcal{R}([a,b]).$$

Then

$$\int_a^b Fg = F(b)G(b) - F(a)G(a) - \int_a^b fG.$$

Proof. Consider FG, which is differentiable on (a,b), continuous on $[a,b]$, with

$$(FG)' = F'G + FG' = fG + Fg.$$

Thus

$$\begin{array}{ccc} \int_a^b (FG)' & = & \int_a^b (fG + Fg) \\ \| & & \| \\ (FG)(b) - (FG)(a) & = & \int_a^b fG + \int_a^b Fg. \end{array}$$

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$$\implies \quad \int_a^b Fg = F(b)G(b) - F(a)G(a) - \int_a^b fG.$$

□

Corollary (Taylor's theorem with Integral Form of Remainder)

Let $x, x_0 \in \mathbb{R}$, $x \neq x_0$,

$$I = \overline{[x, x_0]}, \qquad \mathring{I} = (x, x_0),$$

$f : I \to \mathbb{R}$ be n-times continuously differentiable on I, $n+1$ times differentiable on $\mathring{I}$ such that the function

$$t \longmapsto f^{(n+1)}(t)(x-t)^n \in \mathcal{R}([a,b]).$$

Then

$$(*) \qquad f(x) = \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(x-x_0)^k + \frac{1}{n!} \int_{x_0}^{x} f^{(n+1)}(t)(x-t)^n \, dt,$$

where

$$\int_x^{x_0} = -\int_{x_0}^{x} \quad \text{if } x_0 < x.$$

(if $x_0 < x$, $I = [x_0, x]$ and $\mathring{I} = (x_0, x)$).

Proof. By induction: By the Fundamental Theorem of Calculus part 2, we have

$$f(x) - f(x_0) = \int_{x_0}^{x} f'.$$

Thus

$$f(x) = f(x_0) + \int_{x_0}^{x} f'.$$

By integration by parts,

$$\begin{aligned}\int_{x_0}^{x} f' &= -\left(\int_{x_0}^{x} f'(t)(x-t)'\,dt\right)\\ &= -\left[f'(x)(x-x) - f'(x_0)(x-x_0) - \int_{x_0}^{x} f''(t)(x-t)\,dt\right]\\ &= f'(x_0)(x-x_0) + \int_{x_0}^{x} f''(t)(x-t)\,dt.\end{aligned}$$

Thus

$$f(x) = \sum_{k=0}^{n} \frac{1}{k!} f^{(k)}(x_0)(x-x_0)^k + \frac{1}{n!}\int_{x_0}^{x} f^{(n+1)}(t)(x-t)^n\,dt, \qquad n = 0, 1.$$

Inductive Step: Suppose we have proven $(*)$ for some $m \in \{1, \dots, n-1\}$.

Notice that integration by parts still holds if we have

$$Fg,\ fG \in \mathcal{R}([a,b])$$

by the exact proof; $\underbrace{f}_{F'}, \underbrace{g}_{G'} \in \mathcal{R}([a,b])$ is too strong to ask for.

By "f", "g" we mean the notation in integration by parts, $\neq f$ in this theorem's proof.

Then with

$$\frac{1}{m!}\int_{x_0}^{x} f^{(m+1)}(t)(x-t)^m\,dt,$$

choose

$$\widetilde{F} := f^{(m+1)}(t), \qquad \widetilde{G} := -\frac{1}{m+1}(x-t)^{m+1}.$$

$$\widetilde{F}\widetilde{g},\ \widetilde{f}\widetilde{G} \in \mathcal{R}(I):$$

If $m < n-1$: $f^{(m+2)}$ is continuous, hence Riemann integrable, $f^{(m+1)}$ is also continuous, hence Riemann integrable. On the other hand, polynomials are always continuous, hence Riemann integrable

$$\Longrightarrow \quad \text{“}\widetilde{F}\widetilde{g}\text{”},\ \text{“}\widetilde{f}\widetilde{G}\text{”} \in \mathcal{R}([a,b]).$$

If $m = n-1$: $f^{(m+1)}$ is still continuous, and by assumption,

$$\text{“}fG\text{”} = f^{(n+1)}(t)\frac{(x-t)^n}{-n} \in \mathcal{R}([a,b]) \Longrightarrow \widetilde{F}\widetilde{g} \in \mathcal{R}([a,b]).$$

Therefore by integration by parts, since

$$\frac{1}{m!}\int_{x_0}^{x} f^{(m+1)}(t)(x-t)^m\,dt$$

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$$\begin{aligned}&= -\frac{1}{m!}\left[f^{(m+1)}(x)\frac{(x-x)^{m+1}}{m+1} - f^{(m+1)}(x_0)\frac{(x-x_0)^{m+1}}{m+1}\right.\\ &\qquad\qquad \left. - \int_{x_0}^{x} f^{(m+2)}(t)\frac{(x-t)^{m+1}}{m+1}\,dt\right]\\ &= \frac{1}{(m+1)!}f^{(m+1)}(x_0)(x-x_0)^{m+1} + \frac{1}{(m+1)!}\int_{x_0}^{x} f^{(m+2)}(t)(x-t)^{m+1}\,dt.\end{aligned}$$

Hence $(*)$ holds with $m+1$.

Therefore by induction, $(*)$ holds for $0 \le m \le n$.

Thus $(*)$ holds for n (the case we care about). □

8.2 Improper Riemann–Stieltjes Integrals

Definition

Let $a \in \mathbb{R}$, $b \in (a, \infty]$, and $\alpha : [a, b) \to \mathbb{R}$ increasing. We say that

$$f : [a, b) \to \mathbb{R}$$

is Riemann–Stieltjes integrable over $[a, b)$ with respect to α if

$$f \in \mathcal{R}([a, s], \alpha), \qquad \forall s \in (a, b),$$

and

$$\lim_{s \to b} \int_a^s f \, d\alpha$$

exists (properly, that is, $< \infty$). We denote it

$$\int_a^b f \, d\alpha \qquad \text{or} \qquad \int_a^{b^-} f \, d\alpha \quad \text{if } b < \infty.$$

Thus

$$\int_a^b f \, d\alpha = \lim_{s \to b} \int_a^s f \, d\alpha.$$

Let $b \in \mathbb{R}$, $a \in [-\infty, b)$, and $\alpha : (a, b] \to \mathbb{R}$ increasing. We say that

$$f : (a, b] \to \mathbb{R}$$

is Riemann–Stieltjes integrable over $(a, b]$ with respect to α if

$$f \in \mathcal{R}([s, b], \alpha), \qquad \forall s \in (a, b),$$

and

$$\lim_{s \to a} \int_s^b f \, d\alpha$$

exists (properly, that is, $< \infty$). We denote it

$$\int_a^b f \, d\alpha \qquad \text{or} \qquad \int_{a^+}^b f \, d\alpha \quad \text{if } b < \infty.$$

Thus

$$\int_a^b f \, d\alpha = \lim_{s \to b} \int_a^s f \, d\alpha.$$

Proposition

Let $[a,b] \subseteq \mathbb{R}$, $\alpha : [a,b] \to \mathbb{R}$ increasing, $f \in \mathcal{R}([a,b],\alpha)$.

(i) If α is continuous at b, then
$$\int_a^{b^-} f\,d\alpha = \int_a^b f\,d\alpha.$$

(ii) If α is continuous at a, then
$$\int_{a^+}^b f\,d\alpha = \int_a^b f\,d\alpha.$$

Remark

* It is not true if α is not continuous at the endpoints!

If α is the dirac function at b:
$$\alpha = \begin{cases} 0 & \text{on } [a,b), \\ 1 & \text{at } b, \end{cases}$$
then
$$\int_a^b d\alpha = 1, \qquad \text{but } \forall s \in (a,b), \quad \int_a^s 1\,d\alpha = 0.$$

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Proof. (i) $\forall s \in (a,b)$,
$$\left|\int_a^b f\,d\alpha - \int_a^s f\,d\alpha\right| = \left|\int_s^b f\,d\alpha\right| \le \int_s^b |f|\,d\alpha.$$
Again, $f \in \mathcal{R}([a,b],\alpha) \Longrightarrow f$ is bounded by definition. Hence
$$\le \int_s^b \sup|f|\,d\alpha$$
$$= \sup|f| \int_s^b d\alpha$$
$$= \sup|f|(\alpha(b) - \alpha(s)) \longrightarrow 0 \quad \text{as } s \to b$$
by continuity of α at b.

(ii) is left as exercise. □

Example

$$\int_1^s \frac{1}{x^p}\,dx \underset{s\to\infty}{=} \begin{cases} \ln s \longrightarrow \infty, & p = 1, \\ \dfrac{1}{p-1}\left(1 - \dfrac{1}{s^{p-1}}\right) \longrightarrow \begin{cases} \dfrac{1}{p-1}, & p > 1, \\ \infty, & p < 1. \end{cases} \end{cases}$$

Hence

$$\int_1^\infty \frac{1}{x^p}\,dx = \frac{1}{p-1}, \qquad p > 1;$$

$$f(x) = \frac{1}{x^p} \text{ is not Riemann integrable on } [1,\infty), \quad p \le 1.$$

Example

$$\int_s^1 \frac{1}{x^p}\,dx \underset{s\to 0}{=} \begin{cases} -\ln s \longrightarrow \infty, & p = 1, \\ \dfrac{1}{1-p}\left(1 - \dfrac{1}{x^{p-1}}\right) \longrightarrow \begin{cases} \dfrac{1}{1-p}, & \text{if } p < 1, \\ \infty, & \text{if } p > 1. \end{cases} \end{cases}$$

Hence

$$\int_0^1 \frac{1}{x^p}\,dx = \frac{1}{1-p}, \qquad \text{for } p < 1;$$

$$f(x) = \frac{1}{x^p} \text{ is not Riemann integrable on } (0,1] \text{ if } p \ge 1.$$

(*iii*) Assignment 6, Question 4:

$$\int_0^1 x^p\,d\alpha, \qquad \alpha(x) = \left\lfloor -\frac{1}{x} \right\rfloor.$$

Then we can compute

$$\int_s^1 x^p\,d\alpha = \sum_{k=1}^{\lfloor -1/s \rfloor} \frac{1}{k^p}, \qquad \text{for } s \in (0,1),$$

and then let $s \to 0$.

Remark

We only proved

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

on $(0,2)$ where

$$\ln x = \sum_{k=1}^{\infty} \frac{(-1)^{k-1}(x-1)^k}{k}.$$

To have the first example above working, we shall prove:

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Proposition

$$(\ln x)' = \frac{1}{x}, \qquad \forall x \in (0,\infty).$$

Proof. By property of exponentials we proved,

$$e^{\ln x + \ln y} = e^{\ln x} e^{\ln y} = x \cdot y \quad \Longrightarrow \quad \ln x + \ln y = \ln xy.$$

Furthermore,

$$\ln x + \ln \frac{1}{x} = \ln 1 = 0 \quad \Longrightarrow \quad \ln\left(\frac{1}{x}\right) = -\ln x.$$

Thus $\forall x \in (0,\infty)$, $0 < t \le x$,

$$\begin{aligned}\frac{1}{t}(\ln(x+t) - \ln(x)) &= \frac{1}{t}\ln\left(1+\frac{t}{x}\right) \\ &= \frac{1}{t}\sum_{k=1}^{\infty}\frac{(-1)^{k-1}}{k}\left(\frac{t}{x}\right)^k \\ &= \sum_{k=1}^{\infty}\frac{(-1)^{k-1}}{k}\frac{t^{k-1}}{x^k}.\end{aligned}$$

By the proposition on differentiability of power series,

$$\sum_{k=1}^{\infty}\frac{(-1)^{k-1}}{k}\frac{t^{k-1}}{x^k} \quad \text{for } t < |x|$$

is differentiable on that interval, hence continuous. Since it equals 0 at $t = 0$, it follows

$$\frac{d}{dx}(\ln x) = \lim_{t\to 0}\sum_{k=1}^{\infty}\frac{(-1)^{k-1}}{k}\frac{t^{k-1}}{x^k} \overset{\text{continuity}}{=} \sum_{k=1}^{\infty}\frac{(-1)^{k-1}}{k}\frac{0^{k-1}}{x^k} = \frac{1}{x}.$$

□

8.3 Integral Convergence Test

Theorem (Integral convergence test)

Let $n_0 \in \mathbb{N}$, $f : [n_0,\infty) \to [0,\infty)$ be decreasing,

$$\lim_{x\to\infty} f(x) = 0, \qquad f \in \mathcal{R}([n_0, n]), \quad \forall n > n_0.$$

Denote

$$s_n = \sum_{k=n_0}^{n} f(k), \qquad t_n = \int_{n_0}^{n} f, \qquad d_n = s_n - t_n.$$

Then

(i)
$$0 \le f(n+1) \le d_{n+1} \le d_n \le f(n_0).$$

(ii)
$$d_n \longrightarrow d \quad \text{as } n \to \infty \quad \text{for some } d \in [0, f(n_0)].$$

(iii)
$$\sum_{k=n_0}^{\infty} f(k) \text{ converges} \iff f \text{ is Riemann integrable on } [n_0,\infty).$$

(iv)
$$0 \le d_n - d \le f(n), \qquad \forall n > n_0.$$

Proof. (i)

$$\begin{aligned} d_{n+1} &= \sum_{k=n_0}^{n+1} f(k) - \int_{n_0}^{n+1} f \\ &= \sum_{k=n_0}^{n+1} f(k) - \sum_{k=n_0}^{n} \int_k^{k+1} f \\ &= f(n+1) + \sum_{k=n_0}^{n} \left(f(k) - \int_k^{k+1} f \right) \\ &= f(n+1) + \sum_{k=n_0}^{n} \int_k^{k+1} (f(k) - f) \\ &\geq f(n+1) + 0 = f(n+1), \\ &\qquad \underbrace{f(k) - f}_{\geq 0,\ \forall t \in [k,k+1]} \qquad \text{by decreasing.} \end{aligned}$$

$$\begin{aligned} d_n &= \sum_{k=n_0}^{n} f(k) - \int_{n_0}^{n} f \\ &= \sum_{k=n_0+1}^{n} f(k) + f(n_0) - \sum_{k=n_0+1}^{n} \int_{k-1}^{k} f \\ &= f(n_0) + \sum_{k=n_0+1}^{n} \int_{k-1}^{k} (f(k) - f) \\ &\leq f(n_0) + 0 = f(n_0), \\ &\qquad \underbrace{f(k) - f}_{\leq 0,\ \forall t \in [k-1,k]} . \end{aligned}$$

$$d_{n+1} - d_n = f(n+1) - \int_n^{n+1} f = \int_n^{n+1} (f(n+1) - f) \leq 0,$$
$$\underbrace{f(n+1) - f}_{\leq 0,\ \forall t \in [n,n+1]}.$$

(ii) follows by the monotone convergence theorem.

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(iii) Since $d_n \to d$,

$$d_n = s_n + t_n,$$

$$\sum_{k=n_0}^{\infty} f(k) \text{ converges} \iff s_n \text{ converges} \iff t_n \text{ converges} \iff * \int_{n_0}^{\infty} f \text{ is Riemann integrable.}$$

$$\| \qquad \int_{n_0}^{n} f$$

(iv) $d_{n+1} \leq d_n$ by (i). Hence $d_n - d \geq 0$.

Then $\forall n < n' \in \mathbb{N}$,

$$\begin{aligned} d_n - d_{n'} &= \sum_{k=n_0}^{n} f(k) - \sum_{k=n_0}^{n'} f(k) - \left(\int_{n_0}^{n} f - \int_{n_0}^{n'} f \right) \\ &= \int_{n}^{n'} f - \sum_{k=n+1}^{n'} f(k) \\ &= \sum_{k=n+1}^{n'} \int_{k-1}^{k} f - \sum_{k=n+1}^{n'} f(k) \\ &\leq \sum_{k=n+1}^{n'} (f(k-1) - f(k)) \\ &= f(n) - f(n') \\ &\leq f(n). \end{aligned}$$

Sending $n' \to \infty$ gives the result. □

Example

$$f(x) = \frac{1}{x^p}, \qquad p > 0.$$

$$s_n = \sum_{k=1}^{n} \frac{1}{k^p}, \qquad t_n = \int_{n_0}^{n} \frac{1}{x^p}\, dx.$$

(Assignment 5, Question 2 proved this too.)

$\frac{1}{x^p}$ is Riemann integrable over $[1, \infty)$ if and only if $p > 1$. Hence

$$\sum_{k=1}^{\infty} \frac{1}{k^p} \text{ converges if and only if } p > 1.$$

Moreover,

$$0 \leq \sum_{k=1}^{n} \frac{1}{k^p} - \int_{1}^{n} \frac{1}{x^p}\, dx - d \leq \frac{1}{n^p}, \qquad \text{for some } d \in [0, 1].$$

In particular, for $p = 1$,

$$0 \leq \sum_{k=1}^{n} \frac{1}{k} - \int_{1}^{n} f - d < \frac{1}{n}.$$

In this case, d is called the Euler's countant, ≈ 0.577.

NOT ON THE EXAMINATION

8.4 Change of Variables and Uniform Limits

Theorem (Change of variable formula)

Let $a, b, A, B \in \mathbb{R}$, $a < b$, $A < B$, $\alpha : [a,b] \to \mathbb{R}$ increasing and $f : [a,b] \to \mathbb{R}$ bounded, $\varphi : [A,B] \to [a,b]$ be strictly increasing and

$$\varphi([A,B]) = [\varphi(A), \varphi(B)] = [a,b].$$

Then

$$f \in \mathcal{R}([a,b],\alpha) \iff f \circ \varphi \in \mathcal{R}([A,B], \alpha \circ \varphi).$$

Moreover, if $f \in \mathcal{R}([a,b],\alpha)$, then

$$\int_a^b f \, d\alpha = \int_A^B f \circ \varphi \, d(\alpha \circ \varphi).$$

Furthermore, if α and φ are differentiable, then

$$\int_a^b f \, d\alpha = \int_A^B (f \circ \varphi)(\alpha \circ \varphi)' \, dx.$$

Proof. Notice that φ increasing, $\varphi([A,B]) = [a,b] \implies \varphi$ is bijective.

Then observe:

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if P is a partition of $[A,B]$, $\varphi(P)$ is a partition of $[a,b]$ by bijectivity.

Hence

$$\begin{aligned} U(P, f \circ \varphi, \alpha \circ \varphi) &= \sum_{i=1}^{n} \sup_{x_{i-1} \le x \le x_i} (f \circ \varphi) \cdot \Delta(\alpha \circ \varphi)_i \\ &= \sum_{i=1}^{n} \sup_{\varphi(x_{i-1}) \le x \le \varphi(x_i)} f \cdot [\alpha \circ \varphi(x_i) - \alpha \circ \varphi(x_{i-1})] \\ &= U(\varphi(P), f, \alpha). \end{aligned}$$

Likewise,

$$L(P, f \circ \varphi, \alpha \circ \varphi) = L(\varphi(P), f, \alpha).$$

Then all the claims except the last one follows by applying the Riemann integrability criterion to

$$U(\varphi(P), f, \alpha) - L(\varphi(P), f, \alpha) = U(P, f \circ \varphi, \alpha \circ \varphi) - L(P, f \circ \varphi, \alpha \circ \varphi).$$

Lastly

$$\int_a^b f \, d\alpha = \int_A^B (f \circ \varphi)(\alpha \circ \varphi)' \, dx$$

follows from

$$\int_a^b f \, d\alpha = \int_A^B f \circ \varphi \, d(\alpha \circ \varphi)$$

and a question in Assignment 6. □

Proposition

Let $[a,b] \subseteq \mathbb{R}$, $\alpha : [a,b] \to \mathbb{R}$ be increasing and $\{f_k\}_{k\in\mathbb{N}}$ be a sequence in $\mathcal{R}([a,b],\alpha)$ that converges uniformly to some function f on $[a,b]$. Then

$$f \in \mathcal{R}([a,b],\alpha) \qquad \text{and} \qquad \lim_{k\to\infty} \int_a^b f_k \, d\alpha = \int_a^b f \, d\alpha.$$

Proof. Exercise. □